HARDWARE AND SYSTEMS

Hardware and low-level systems from first principles

Handbook

How the course works, what to buy, how to set up the bench, and how to track your progress

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PART 1

The course

Hardware and Systems is a lab-driven course in electronics and low-level systems for one student: a strong software engineer who is founding OMH, a company that will design, manufacture and support rack modules delivering cloud-class services on its own hardware. It starts with electrons and ends with two working systems that you build from the transistor up. The first is a quadcopter that flies on your firmware, your boards, your radio protocol and a TI radar altimeter. The second is Module Zero, a two-node fleet of engineering-sample server modules that attest their firmware, form a quorum, run tenant virtual machines on your hypervisor and store data erasure-coded across NVMe. For every claim on the OMH hardware page (zero-touch join, measured boot, OpenBMC and Redfish, rolling firmware updates, encryption with keys held in Modulus), you will have built a small working version.

The course is built around specific Texas Instruments silicon wherever TI makes the part: precision sensors (TMP117, HDC3022, INA228, TMAG5273, OPT3101, OPT4048), converters (ADS1115, ADS1220), power (TPS62933, TPS25947, LM74700), microcontrollers (MSP432E401Y, MSPM0G3507, C2000 F280025C), motor control (DRV8323RS), sub-gigahertz radio (CC1312R), microwave synthesis (LMX2572) and millimetre-wave radar (IWRL6432). Where TI makes nothing suitable, such as MEMS inertial sensors, the course says so and names the part it uses instead.

What you will have built

- A CMOS NAND gate from discrete transistors, a 4-bit adder from logic ICs, and then a pipelined RV32I CPU with caches and M-mode traps on an FPGA, running C you compiled.
- Library-free drivers and metrology for five TI sensors and two TI converters, an error budget for a real measurement, characterised TI power stages and protection, and a characterised drone battery.
- A transmission-line measurement, a tuned antenna, an FM receiver written from samples, and link budgets checked in the field.
- Bare-metal firmware for TI's MSP432E401Y with no vendor library: your own linker script, startup code, drivers, fault handler and a signed A/B bootloader with anti-rollback.
- A boot chain you understand byte by byte: a boot sector, a UEFI loader, Secure Boot with your own keys, measured boot into a TPM, remote attestation with a nonce, zero-touch netboot, and OpenBMC.
- Field-oriented motor control written from scratch on a C2000, an IMU noise model, an attitude estimator and a PID controller tuned on a rig.
- A packet radio written below TI's protocol stacks with a measured sensitivity waterfall, a telemetry protocol designed to a duty-cycle budget, and a software demodulator.
- A frequency synthesiser programmed from your own register calculator, a measured chirp, an FMCW radar simulator, and a TI 60 GHz radar configured as an altimeter.
- Four printed circuit boards you specified, designed, reviewed, had fabricated and brought up: the drone's power module and flight-controller board, a BMC-lite management controller on the MSP432E401Y, and a Compute Module 5 carrier with PCIe NVMe and a TPM.
- A 450 mm quadcopter that hovers, holds altitude on radar and barometer, and reports telemetry over your 868 MHz link.
- A 64-bit SMP kernel with user mode and an NVMe driver; Linux drivers for TI parts that mainline Linux lacks; a KVM virtual machine monitor with your own virtio devices; storage and network data paths measured from `io_uring` to SPDK and from the kernel stack to DPDK.
- A written product specification, threat model, update architecture, manufacturing test plan, compliance plan and cost model, and the Module Zero demonstration that ties it together.

Four units

Unit	Chapters	Weeks
1 Circuits, Sensors and Signals	Electricity, components and measurement; the analogue signal chain; sensors with TI silicon; power conversion and batteries; signals on wires and RF fundamentals	11.5
2 Logic, Processors and Firmware	Digital logic; HDL and FPGA; build a CPU; memory and interconnect; privilege and the platform; bare-metal firmware; boot and platform firmware	21
3 Control, Radio and Boards	Motors, estimation and feedback control; radio links; microwave synthesis and FMCW radar; board design and bring-up; the drone	21, plus fabrication
4 Systems Software and the Product	Write a kernel; Linux kernel internals; Linux device drivers; virtualisation and hypervisors; storage and network data paths; product engineering; the Module Zero capstone	about 29

The weeks assume 12 to 15 hours a week: two weekday evenings of reading and problem sets and one long session on the bench or in the debugger. The whole course takes about eighty weeks. The week counts are a guide, and the done-when criteria decide when a piece of work is finished.

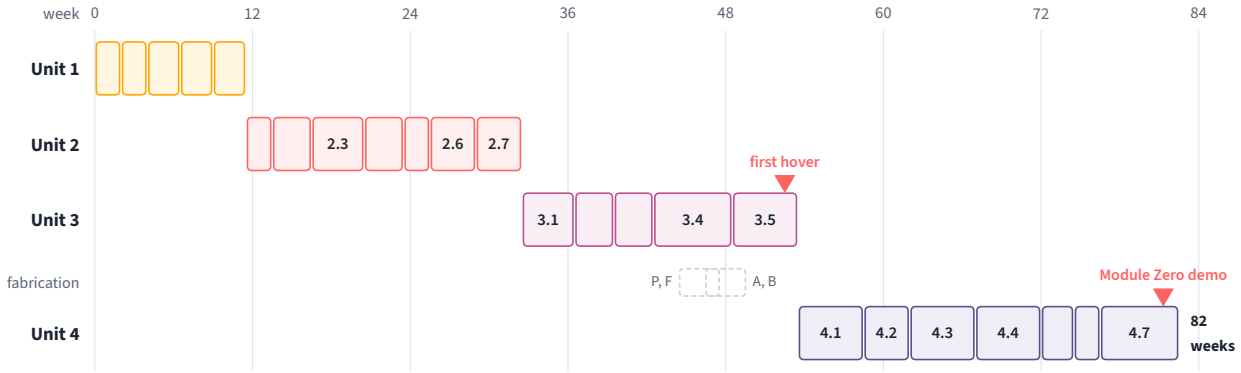


Figure 1.1. The course as a schedule at 12 to 15 hours a week. The two board orders in Chapter 3.4 are in fabrication while you work on the rest of the chapter and the first, software-only labs of Chapter 3.5.

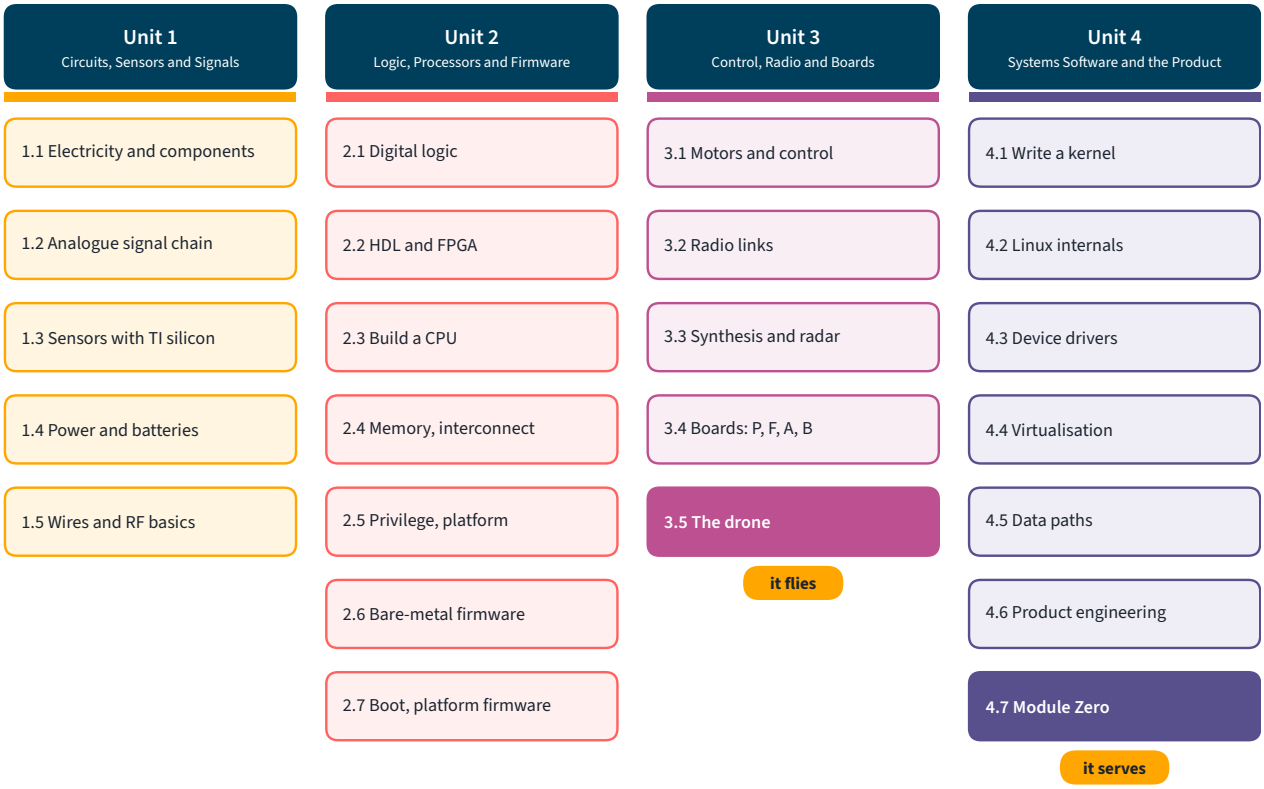


Figure 1.2. The course map. Each unit builds on the units before it. The threads end in the drone (Chapter 3.5) and Module Zero (Chapter 4.7).

The threads

Six threads run through the units, and most chapters advance more than one.

- **Power.** Ohm's law and decoupling (1.1), converters, protection and batteries (1.4), the power trees that become Boards P and A (3.4), the drone's energy budget (3.5) and Module Zero's power telemetry (4.7).
- **Sensing and measurement.** The signal chain (1.2), TI sensors and error budgets (1.3), the IMU and magnetometer (3.1), radar (3.3), and the hwmon and IIO drivers that put them in Linux (4.3).
- **Compute and firmware.** Logic to a CPU (2.1 to 2.4), privilege (2.5), bare-metal firmware (2.6), the flight software (3.5), your kernel (4.1) and Linux (4.2).
- **Trust.** The signed bootloader (2.6), Secure Boot, measured boot and attestation (2.7), the BMC (2.7, 3.4), the threat model (4.6) and Module Zero's join protocol (4.7).
- **Radio and high-speed signals.** Transmission lines and link budgets (1.5), the 868 MHz link (3.2), synthesis and radar (3.3), and PCIe and Ethernet routing on Board B (3.4).
- **The product.** Supply chains and kit (this handbook), design reviews and respins (3.4), flight-test reports (3.5), and the specification, compliance and cost documents (4.6).

Order and flexibility

The units run in order, and so do the chapters within them, with four sanctioned exceptions. Board A's specification (Lab 3.4.1) can start as soon as Chapter 2.6 is done, so that fabrication lead time stays off the critical path. Chapters 3.2 and 3.3 do not depend on 3.1 and can run in either order. Chapter 4.5 can follow 4.3 directly. Product engineering (4.6) can be read alongside the capstone's first two weeks.

Conventions

Spelling is British. “The Pi” is the Raspberry Pi 5, “the server” is the used x86 rack server from the kit, “the FPGA” is the ULX3S, and “the LaunchPad” is the MSP-EXP432E401Y unless a lab names another. “The Windows PC” is the small x86 machine that runs TI’s Windows-only tools. Your workstation is assumed to be a Mac with Apple silicon. Commands on the Pi and the server assume Raspberry Pi OS and Debian or Ubuntu. Prices are those shown by the named seller on 4 October 2026. Labs are numbered unit.chapter.lab, so Lab 2.6.6 is the sixth lab of Chapter 2.6.

PART 2

How to work

The anatomy of a chapter

Every chapter has the same parts in the same order, so you always know where you are.

- **The strip under the title** gives the time the chapter takes, what you will have built at the end, and exactly what it needs from the kit.
- **What this buys OMH** says why the chapter is in a course for a server company. Read it first, because it tells you what to pay attention to.
- **The skip test** is five questions. If you can answer all five cold, in writing, and would be comfortable being questioned on them, its rule tells you which labs to do, and you can skim the rest. Answer honestly, because most of the learning happens in the labs.
- **Core ideas** is the theory you need for the labs, with figures. It is deliberately dense, and each paragraph covers something you will use.
- **Reading** names specific chapters and sections. Read them before the lab that needs them.
- **Labs** make up most of the course. Each has a goal, a time estimate and a kit line, then numbered steps, then **done when** criteria and an **evidence** list.
- **The problem set** checks the theory with numbers. Answer every problem in writing with units, and check each numerical answer a second way.
- **Deliverables and stretch** list what goes into your labs repository, and what to do if you have time to spare.
- **The completion checklist** has one line per lab, one for the problem set and one for what you can explain without notes.

Each unit ends with a **sign-off** page: a table of its chapters and a set of review tasks that test whether you can connect the unit's ideas.

The completion standard

A lab is done when every **done when** line holds and every item in its evidence list exists. A chapter is done when every line of its checklist is ticked. A unit is done when its sign-off page is complete. Do not move on with a line unticked. If something outside your control blocks a line (a part out of stock, a board at fabrication), write the reason and the date in the notebook, mark the line as deferred, and come back to it. A deferred line does not count as ticked.

Each criterion is a measurable statement, such as "the measured corner is within 5 % of the design and the plot overlays the simulation". A statement such as "the filter works" is too vague to check. You may meet a criterion by a route the lab did not describe, provided you write down what you did.

The lab notebook

Keep one Markdown file per chapter in your labs repository. For every lab, record:

1. the date and the goal in one line;
2. the set-up, with a photograph and the instrument settings;
3. your prediction before you measure, with the arithmetic;
4. the measurement, with raw data saved as files and named in the entry;
5. the result against the prediction, and an explanation of every discrepancy larger than your stated uncertainty;

6. what you would do differently.

Scope captures go in `captures/`, raw data as CSV in `data/`, and every plot is produced by a script in `scripts/` that reads the raw data, so you can regenerate it. The notebook is also the raw material for OMH's engineering blog, through which a hardware company with no customers can earn credibility.

```
## Lab 1.2.2  A second-order filter and its Bode plot   2026-11-14
Goal: Sallen-Key low-pass near 1 kHz, measured against its design.
Set-up: TLV2372IP on 5 V, R = 11 kΩ, C1 = 22 nF, C2 = 10 nF (measured below).
Prediction: fc = 1/(2π R √(C1 C2)) = 976 Hz, Q = 0.74; -40 dB/decade above.
Measured: data/bode-2026-11-14.csv; fc = 951 Hz (-2.6 %).
Discrepancy: C1 measures 22.9 nF; recomputed fc = 956 Hz, within 0.6 %.
Next time: measure every capacitor before building.
```

The labs repository

Create one Git repository for the coursework, with a directory per chapter and a short README in each that links that chapter's evidence.

```
omh-labs/
  u1/1.1-electricity/    notebook.md  captures/  data/  scripts/  sim/
  u1/1.3-sensors/        notebook.md  sensors/   data/  scripts/
  u2/2.2-fpga/           rtl/  tb/  formal/  constraints/
  u2/2.6-firmware/       firmware/  bootloader/  tools/
  u3/3.2-radio/          firmware/  ground/  sdr/
  u3/3.4-boards/         board-p/  board-f/  board-a/  board-b/  reviews/
  u3/3.5-drone/          flight/  sim/  logs/  test-cards/
  u4/4.3-drivers/        opt4048/  ads1220/  emc2101/  fpga-spi/
  u4/4.7-module-zero/    design-notes/  image/  control-plane/
```

Commit small and often, with messages that say what changed and why. Tag the commit that completes each chapter (`ch-2.6-done`), so the sign-off table can point at it.

Habits

Read the datasheet before any tutorial. When a lab involves a chip, read its datasheet or reference manual chapter first and only then look at anyone else's code. OMH's engineers will routinely work from primary sources alone, for parts that have no tutorials.

Write your own version before you use an abstraction. Chapter 2.6 forbids TI's driverlib, Chapter 2.3 comes before any SoC generator, and your `/dev/kvm` program comes before kvmtool. Two chapters allow vendor code for set-up only and say so: C2000Ware's driverlib for peripheral initialisation in Chapter 3.1, and TI's RF driver with SysConfig settings in Chapter 3.2. Afterwards, use whatever you like and read it critically.

Predict, then measure. Write the number you expect before you look at the instrument. Without a prediction, a measurement cannot test your understanding.

Treat the instrument as part of the measurement. Chapters 1.1, 1.5 and 3.3 each include a measurement in which the instrument gives a misleading result. Learn an instrument's limits before you rely on its readings.

Debug by bisection. Reproduce the failure, make it smaller, measure at the midpoint, and compare with what the datasheet says should be there. When you ask for help, attach the evidence. TI's E2E forums, the EEVblog forum and the kernel mailing lists all answer questions that show the work.

Order early and keep a supply-chain log. Note what you ordered, when, the lead time quoted and the lead time delivered. Chapter 4.6's cost model starts from that log.

PART 3

Safety on the bench

By design, nothing in this course works at dangerous voltages. The hazards are stored energy (lithium-polymer packs and capacitors), heat (soldering irons, hot air and hot components), spinning propellers, and the mains equipment on the bench. The rules below apply to every chapter; individual labs add their own.

Mains and the bench

- The only things plugged into the mains are bought, closed instruments and chargers: the bench supply, the electronic load, the oscilloscope, the generator, the soldering station, the charger, the computers. Never open one, and never build anything that connects to the mains.
- Use sockets protected by a residual-current device, and keep liquids off the bench.
- Set the bench supply's voltage and current limit before you connect anything, with the output off. For a first power-on, limit the current to a fraction of what you expect the circuit to draw.
- Check polarity with the multimeter before connecting any supply to a board. Most of the TI EVMs survive a reversed supply, but most of your boards will not.
- Large capacitors hold their charge after power is removed. Discharge them through a resistor and confirm with the meter before you touch the circuit.

Static

Work on the anti-static mat with the wrist strap connected, and keep boards in their bags when not in use. The radar, the synthesiser and the radio boards are the most sensitive in the kit. ESD damage can be partial: the board still works but performs slightly worse, which ruins a measurement without any visible sign.

Soldering, rework and chemicals

- A soldering tip runs at 300 to 400 °C and a hot-air nozzle at up to 500 °C. Return the iron to its stand every time you put it down, keep the hot-air handle in its cradle so it switches off, and let parts cool before you touch them.
- Hot air melts plastic connectors and lifts nearby parts; shield them with Kapton tape and practise on a scrap board first.
- Flux fumes irritate the lungs, so work with a window open or a fume extractor drawing the plume away from your face.
- Leaded solder is easier to learn on and is used in this course for that reason. Wash your hands after soldering, do not eat at the bench, and keep the solder away from children. Anything OMH ships will use lead-free solder (Chapter 4.6).
- Isopropyl alcohol is flammable. Keep it closed and away from the iron and hot air.
- Wear safety glasses when you clip leads, because the offcuts fly off.

Lithium-polymer packs

The drone's 3S pack stores about 55 Wh and can deliver over 100 A into a short circuit. It is the most dangerous object in the kit. The British Model Flying Association's battery booklet is the basis for these rules.

- **Charging.** Use the balance charger in balance mode, constant current then constant voltage, to no more than 4.20 V per cell, at 1C (5 A for the 5,000 mAh pack) unless the maker allows more. Charge inside the LiPo bag on a non-flammable surface, never when the pack is hot or below 5 °C, and never unattended, while you sleep, or in an escape route.
- **Discharging.** Never go below 2.8 V per cell. Set every cut-off at 3.0 to 3.3 V per cell under load. This course lands the drone at 3.5 V per cell.
- **Storage.** Store at 3.6 to 3.8 V per cell (about 60 % charge), at 10 to 25 °C, in a vented metal box or the LiPo bag, never above 45 °C or in direct sunlight. The charger's storage mode does this.
- **Damage.** A pack that is puffed, hot, dented or crash-damaged must never be used again. Stop, disconnect, and move it to the fire-safe container outdoors if you can do so safely. Do not open it. Once it is cool and stable, discharge it slowly through a resistive load (never by shorting the leads), tape the connectors, label it, and take it to a household waste recycling centre.
- **Fire.** A lithium fire cannot be put out with a small extinguisher. Get out, close the door, and call the fire service.
- **Shipping.** Couriers restrict lithium batteries; check that the seller delivers them to your address before you order.

Propellers and the test ladder

- Propellers come off for every bench test, firmware flash and wiring change, and go on only for a stand, rig, tether or flight test with a written test card.
- Wear safety glasses for every powered motor test, stand outside the plane of the propellers, and keep other people at least 5 m away.
- Check the kill switch disarms before every test, and test failsafe with propellers off.
- Climb the test ladder one rung at a time (simulation, bench, rig, tether, hover), and step back down when anything is unexplained. Chapter 3.5 describes the ladder in full.
- Fly in an open space over grass with nobody underneath.

RF and microwave

The transmitters in this course are low power: 25 mW at most at 868 MHz and a few milliwatts at 60 GHz. The risk is to the instruments. Fit the 30 dB attenuator directly on a transmitter's output before connecting anything else, compute the level at the instrument before you connect it, and keep the tinySA below 0 dBm. Never connect a transmitter to the network analyser.

The server

A 2U server weighs 20 to 30 kg, so lift it with someone else or put it straight onto rails. It is loud and draws a few hundred watts. Give it airflow, keep its lid on when running (the fans rely on it), and run it where the noise will not stop you working.

PART 4

The kit

Every item the course uses is listed below, by the unit that first needs it, with the exact product, part number, seller and price. The tables are generated from one file, `kit.yaml`, so the course and the shopping list always match. Each item's name links to the page that was checked; the seller links to the same page.

How to read and use the tables

- **Prices** are for one unit, as the seller showed them on 4 October 2026. "inc VAT" prices include UK VAT. "+ VAT" prices (DigiKey UK, some distributors) exclude it, so add 20 %. "+ import VAT" items come from sellers outside the UK, and the courier collects import VAT and a handling fee on delivery. The Isle of Man is inside the UK VAT area, so UK sellers charge VAT as normal.
- **Quantities** are what the course needs. Where a lab consumes parts (an attenuator you might overload, a breakout you might damage), buy the spare the table suggests.
- **Stock.** Several items were back-ordered or sold out when checked; their notes give an alternative or a date. Check stock and price on the day you order.
- **Not verified** marks four items with no fixed listing. The used server and the used enterprise NVMe drive are bought second-hand. Use the checklist later in this part, and expect to pay what a UK refurbisher charges. The small power bank and the isopropyl alcohol are bought locally, because couriers will not easily carry lithium cells or flammable liquids to the Isle of Man.
- **Delivery.** DigiKey UK delivers free above £65 before VAT, so batch DigiKey orders. Lithium batteries and 18650 cells travel by road courier only, so confirm the seller delivers them to you before you order.
- **Optional** items are for stretch work or are alternatives to a required item. They are counted separately in the totals.

When to order

When	What
Before Unit 1	Everything in the Unit 1 table. Also the ULX3S (it ships from the United States), the LP-MSPM0G3507 (back-ordered when checked) and the Raspberry Pi SSD or its alternative (out of stock when checked).
During Unit 1	The rest of the Unit 2 table, including the used server, its NIC and the managed switch: Chapter 2.4 needs the server, and choosing a good one takes time.
During Unit 2	The Unit 3 table except the CM5. TI's motor, radio, synthesiser and radar boards had limited stock (the LVBLDCMTR had three at DigiKey), so order them early. Order the Windows PC before Chapter 3.2.
During Chapter 3.4	Board fabrication and the CM5 module, once Board B has passed its design review.
During Unit 3	The Unit 4 table.

Unit 1 kit

	Required	Optional
UK sellers, VAT included (ex-VAT prices × 1.2)	£3,212	£603
Overseas sellers, before import VAT and delivery	€145	US$26
Items with no checked price	1	1

Not priced: Isopropyl alcohol, 99 % or stronger, bought locally.

Item and use	SKU or part number	Qty	Price each	Seller
Bench instruments and tools				
Brymen BM235 true-RMS multimeter Every lab. Its K-type thermocouple input is the course's reference thermometer. *Ships with gold-tip leads and a K-type probe. 600 µA range reads to 0.1 µA. Shown as back-order.*	`BM235`	1	£87.60 inc VAT	Telonic Instruments
Rigol DHO804 4-channel 12-bit oscilloscope, 70 MHz Every lab from Chapter 1.1 onwards. *1.25 GSa/s, 25 Mpts, USB and LAN for scripted captures. Back-order of one to two weeks. Alternative: Siglent SDS804X HD, £412.80 inc VAT.*	`DHO804`	1	£364.80 inc VAT	Rigol UK
Siglent SPD1305X programmable linear supply, 30 V 5 A Current-limited power for every first power-on; 24 V for the motor labs. *Linear, 4-wire sense, remote control over USB. Needs 4 mm banana leads with crocodile clips.*	`SPD1305X`	1	£259.20 inc VAT	Telonic Instruments
Siglent SDG1032X Plus 2-channel 30 MHz arbitrary waveform generator Filter, ADC and op-amp labs in Chapter 1.2; loop injection in 1.4. *Scriptable over LAN and USB (SCPI), which the Bode-plot lab uses.*	`SDG1032X Plus`	1	£373.20 inc VAT	Siglent UK
Siglent SDL1020X-E programmable DC load, 150 V 30 A 200 W Converter efficiency and transients (1.4), battery discharge curves (1.4.4). *Battery-test mode with cut-off voltage. Needs 30 A leads and an XT60 pigtail for the LiPo.*	`SDL1020X-E`	1	£444.00 inc VAT	Telonic Instruments
ATTEN ST-8802 soldering and hot-air rework station All soldering; hot air for the SMA rework (3.2.1) and board rework (3.4). *65 W iron with T900 tips and an 800 W hot-air handle with four nozzles, UK mains lead. Only one was in stock: if it has gone, buy the Pinecil below for soldering and add hot air later.*	`ST-8802`	1	£189.90 inc VAT	Flying Tech
PINE64 Pinecil V2 USB-C soldering iron OPTIONAL Alternative iron if the ATTEN station is out of stock. *Needs a 20 V USB-C PD supply (the Raspberry Pi 45 W supply, The Pi Hut, £14.40). UK alternative: Miniware TS101 at The Pi Hut, £57.50.*	`Pinecil V2`	1	US$25.99 + import VAT	PINE64
iFixit portable anti-static mat with wrist strap Handling every board, especially the radar and RF boards. *60 × 60 cm mat, 1.8 m earth cord with 10 mm snap, wrist strap.*	`The Pi Hut 102672`	1	£28.00 inc VAT	The Pi Hut
USB logic analyser, 24 MHz, 8 channels (sigrok PulseView) Every bus lab: I²C, SPI, UART, DShot, CRSF. *Inputs to 5.25 V. On some units the ground pin beside input 6 is not ground: use another ground pin.*	`TOL-18627`	1	£26.30 inc VAT	The Pi Hut
Saleae Logic 8 OPTIONAL A faster analyser with analogue capture, if the budget allows. *100 MS/s digital, 10 MS/s analogue. No GBP price could be confirmed.*	`Logic 8`	1	not checked	Saleae, through Mouser
150 mm digital calipers Antenna lengths, footprints, mechanical fits. *Needs an LR44 cell.*	`105194`	1	£9.50 inc VAT	The Pi Hut
iFixit precision tweezers, set of three, ESD-safe Placing and reworking surface-mount parts.	`EU145060`	1	£14.00 inc VAT	The Pi Hut
iFixit Mandible flush cutters Trimming leads and wire. *Copper only; do not cut steel.*	`IF145-548-1`	1	£15.00 inc VAT	The Pi Hut
Topdon TC001 USB-C thermal camera OPTIONAL Finding hot spots and shorts; converter and motor thermal checks. *256 × 192, −20 to 550 °C. Needs an Android phone with USB-C or a Windows PC.*	`TC001`	1	£258.00 inc VAT	Topdon UK
Safety glasses, clear, EN 166 / ANSI Z87+ Every soldering, lead-clipping and powered-motor session. *14 left when checked. The listing states 'ANSI z87+/EN166 compliant'; check for the EN 166 mark on the frame on arrival. The anti-fog version IF145-556-2 is £8 with 18 in stock.*	`IF145-556-1`	2	£4.00 inc VAT	The Pi Hut
Components and consumables				
Full-size 830-point breadboard Units 1 and 2 circuits.	`102817`	2	£5.00 inc VAT	The Pi Hut
Dupont jumper wire pack, 120 pieces Breadboard and module wiring. *40 each of male-male, male-female and female-female. The 140-piece pre-cut kit (£4.00) makes neater breadboards.*	`102394`	1	£6.00 inc VAT	The Pi Hut
Metal-film resistor kit, 575 pieces Units 1 and 2. *16 values from 0 Ω to 1 MΩ. Tolerance not stated: measure any resistor whose value matters.*	`102328`	1	£6.00 inc VAT	The Pi Hut

Item and use	SKU or part number	Qty	Price each	Seller
Electrolytic capacitor kit, 200 pieces RC, decoupling and bulk-capacitance labs. *0.1 to 220 µF, 10 to 50 V.*	`104752`	1	£6.00 inc VAT	The Pi Hut
Ceramic capacitor pack, 250 pieces Decoupling (100 nF) and small-value filters. *25 values from 1 pF to 100 nF.*	`FIT0118`	1	£4.60 inc VAT	The Pi Hut
Chip Quik Sn63/Pb37 solder wire, 0.5 mm, 28 g All soldering. *Leaded solder is easier to learn on. Work with ventilation and wash your hands afterwards.*	`SMDSW.020 1OZ`	1	£5.81 + VAT	DigiKey UK
Future 315 no-clean flux pen Surface-mount work and rework.	`The Pi Hut 106030`	1	£7.80 inc VAT	The Pi Hut
Desoldering braid, 1.5 mm × 1.5 m Removing solder bridges. *Add flux before use.*	`The Pi Hut 104300`	1	£2.60 inc VAT	The Pi Hut
Heat-shrink kit, 265 pieces Insulating joints; XT60 solder cups in 1.4.4 and 3.5. *1.5 to 10 mm diameters.*	`104745`	1	£6.00 inc VAT	The Pi Hut
Noctua NF-A8 PWM 80 mm 12 V 4-pin fan The MOSFET switch load (1.1.3), the INA228 load (1.3.3), and the fan-control labs (2.6, 4.3). *12 V, 4-pin PWM with tachometer, 2,200 rpm.*	`NF-A8 PWM`	1	£16.99 inc VAT	Scan
Bourns RSA-50-50 four-terminal current shunt, 1 mΩ, 50 A High-current measurement with an INA228: the battery lab (1.4.4) and the thrust stand (3.1.1). *Current through the large M4 studs; sense only across the small screws. 2.5 W.*	`RSA-50-50`	1	£34.70 + VAT	DigiKey UK
LM358P dual op-amp, DIP-8 Op-amp stages lab in Chapter 1.2: gain-bandwidth, slew rate, offset and the output-swing limit on 5 V. *Active; 14,195 in stock; £0.17 each at 10.*	`LM358P`	2	£0.24 + VAT	DigiKey UK
TLV2372IP rail-to-rail dual op-amp, DIP-8 Rail-to-rail half of the op-amp lab, the second-order filter and the instrumentation-amplifier stretch in Chapter 1.2. *Active; 2,181 in stock; £1.141 each at 10.*	`TLV2372IP`	2	£1.55 + VAT	DigiKey UK
LM4040AIZ-2.5/NOPB 2.5 V shunt voltage reference, ±0.1 %, TO-92 The known voltage the ADS1115 and the multimeter measure in the sampling lab, Chapter 1.2. *Active; 1,748 in stock; £1.37 each at 10. The A grade (±0.1 %) keeps the reference from being the largest error in the comparison. The D grade LM4040DIZ-2.5/NOPB (±1 %) is £0.49 with 2,718 in stock It is a shunt reference, so it needs a series resistor from the supply; 1 kΩ from 5 V gives about 2.5 mA.*	`LM4040AIZ-2.5/NOPB`	2	£1.85 + VAT	DigiKey UK
LM317KCSE3 adjustable linear regulator, 1.5 A, TO-220 Linear versus switching lab in Chapter 1.4: 5.0 V from 12 V at 300 mA. *Active; 946 in stock; £0.573 each at 10. At 12 V in and 300 mA out it dissipates about 2.1 W and gets hot enough to burn with no heatsink.*	`LM317KCSE3`	2	£0.79 + VAT	DigiKey UK
SN74HC04N hex inverter, DIP-14 Logic thresholds in Chapter 1.1; inverters in Chapter 2.1. *Active; thousands in stock when checked.*	`SN74HC04N`	2	£0.57 + VAT	DigiKey UK
SN74HC14N hex Schmitt-trigger inverter, DIP-14 The decoupling oscillator (1.1.4), the cable driver (1.5.1) and debouncing (2.1). *Active; thousands in stock when checked.*	`SN74HC14N`	4	£0.85 + VAT	DigiKey UK
CSD18534KCS 60 V N-channel MOSFET, TO-220 (fan switch) Low-side switch for the 12 V fan in Chapter 1.1, including the turn-off-spike step without a flyback diode. *Active; 1,511 in stock; £0.955 each at 10. DigiKey lists Vgs(th) at most 2.3 V and Rds(on) at most 9.5 mΩ at a 10 V gate; Rds(on) is not specified at 3.3 V, so measure it in the lab. The Chapter 1.1 MOSFET lab names the IRLZ44N; this TI part takes its place. Fallback: Infineon IRLZ44NPBF, £1.32, 43,047 in stock. No quantity was given; two leaves a spare for the spike step.*	`CSD18534KCS`	2	£1.50 + VAT	DigiKey UK
2N7000 60 V N-channel MOSFET, TO-92 First fan switch in Chapter 1.1; the N-channel half of the discrete CMOS inverter and NAND in Chapter 2.1. *Active; 128,151 in stock. Ten cost £3.02 (£0.302 each), 57p more than five. TI makes no TO-92 MOSFET. Rated 200 mA continuous, so check the fan draws less. TO-92 MOSFETs do not share one pin order; check the datasheet before wiring.*	`2N7000`	5	£0.49 + VAT	DigiKey UK
1N5819 40 V 1 A Schottky diode, DO-41 Flyback diode across the fan in Chapter 1.1; forward-drop comparison with the ideal-diode EVM in Chapter 1.4. *Active; 12,971 in stock; cut tape, sold singly; £0.145 each at 10.*	`1N5819`	5	£0.24 + VAT	DigiKey UK

Item and use	SKU or part number	Qty	Price each	Seller
1N4148 small-signal diode, DO-35 General small-signal diode in the Unit 1 bench kit. *10-piece price (£0.07 singly). Active; 647,904 in stock; sold loose.*	`1N4148`	10	£0.04 + VAT	DigiKey UK
1N4007 1000 V 1 A rectifier diode, axial General rectifier diode in the Unit 1 bench kit. *Active; 381,086 in stock; cut tape, sold singly; £0.064 each at 10.*	`1N4007`	5	£0.10 + VAT	DigiKey UK
5 mm red LED, diffused LED and series-resistor lab in Chapter 1.1; adder outputs in Chapter 2.1. *10-piece price (£0.15 singly). Active; 131,049 in stock. DigiKey lists Vf 1.9 V typical at 10 mA, 627 nm peak, 40 mcd; the Kingbright datasheet is linked from the product page.*	`WP7113ID`	10	£0.11 + VAT	DigiKey UK
5 mm green LED, diffused Indicator LEDs for Chapter 1.1 and the Chapter 2.1 logic labs. *10-piece price (£0.17 singly). Active; 45,121 in stock. DigiKey lists Vf 2.2 V typical at 20 mA, 565 nm peak, 30 mcd; datasheet linked from the product page.*	`WP7113GD`	10	£0.11 + VAT	DigiKey UK
6 × 6 mm tactile push button, through-hole Drives the RC charge lab in Chapter 1.1; the hand clock for the state machine in Chapter 2.1. *Active; 99,187 in stock; 5 mm tall, 160 gf. The four legs are joined in pairs inside; find which pairs with the meter's continuity mode before wiring.*	`1825910-6`	5	£0.10 + VAT	DigiKey UK
11 kΩ 1 % 0.25 W metal-film resistor, axial Second-order filter lab in Chapter 1.2. *10-piece price (£0.07 singly). Active; 6,477 in stock; ±100 ppm/°C.*	`MFR-25FBF52-11K`	10	£0.03 + VAT	DigiKey UK
0.1 Ω 1 % 3 W current-sense resistor, axial Inrush and short-circuit captures in the protection lab, Chapter 1.4. *Active; 25,754 in stock; £2.137 each at 10. Non-inductive metal element. 100 mV across it means 1 A.*	`13FR100E`	2	£3.17 + VAT	DigiKey UK
5 Ω 10 W aluminium-housed wirewound resistor Known load in series with the INA228 shunt in the power-monitor lab, Chapter 1.3. *Active; 110 in stock; £3.69 each at 10. ±5 %, solder-lug terminals. The 10 W rating assumes it is bolted to a heatsink; in free air allow roughly half.*	`HS10 5R J`	2	£5.39 + VAT	DigiKey UK
1,000 µF 25 V low-ESR aluminium electrolytic capacitor, radial The capacitor hot-plugged onto 12 V in the inrush lab, Chapter 1.4. *Active; 1,533 in stock; £0.671 each at 10. FR series: DigiKey lists 20 mΩ, 2.18 A ripple at 100 kHz, 10,000 h at 105 °C, 10 mm diameter, 5 mm lead spacing. Polarised: the stripe marks the negative lead.*	`EEU-FR1E102`	2	£1.05 + VAT	DigiKey UK
22 nF 50 V ±5 % C0G ceramic capacitor, radial Second-order filter lab in Chapter 1.2. *Active; 11,838 in stock; £0.261 each at 10. 5 mm lead spacing.*	`FG24C0G1H223JNT06`	5	£0.43 + VAT	DigiKey UK
10 nF 50 V ±5 % C0G ceramic capacitor, radial Second-order filter lab in Chapter 1.2. *Active; 27,075 in stock; £0.167 each at 10. 5 mm lead spacing.*	`FG28C0G1H103JNT06`	5	£0.28 + VAT	DigiKey UK
STEMMA QT / Qwiic JST-SH 4-pin cable, 100 mm Daisy-chaining the I2C sensor breakouts in Chapter 1.3. *In stock. The Pi Hut's own-brand cable, so the SKU is The Pi Hut's; no manufacturer part number. Female JST-SH on both ends; red 3.3 V, black GND, blue SDA, yellow SCL. SparkFun's PRT-14427 was sold out.*	`104221`	5	£1.00 inc VAT	The Pi Hut
STEMMA QT / Qwiic JST-SH 4-pin cable, 200 mm Longer I2C runs in Chapter 1.3, such as the magnet rig. *In stock. The Pi Hut's own-brand cable; the SKU is The Pi Hut's.*	`104965`	2	£1.00 inc VAT	The Pi Hut
STEMMA QT / Qwiic JST-SH 4-pin to male jumper wires, 150 mm Connecting a Qwiic breakout to the breadboard or the logic analyser in Chapter 1.3. *In stock. The Pi Hut's own-brand cable; the SKU is The Pi Hut's.*	`104964`	2	£1.00 inc VAT	The Pi Hut
Neodymium disc magnet 6 × 2.5 mm, diametrically magnetised The rotating magnet above the TMAG5273 in the magnetic-angle lab, Chapter 1.3. *Seller substitution: first4magnets refused the fetch (HTTP 403) and e-Magnets UK had only a 5 × 10 mm diametric rod in packs of 20 (£16.94), so this comes from DigiKey UK with the rest of the order. N35, NiCuNi plated; active; 30,023 in stock; £0.244 each at 10. Keep magnets away from small children.*	`8995`	2	£0.29 + VAT	DigiKey UK
Isopropyl alcohol, 99 % or stronger, bought locally Removing flux after soldering and rework. *No UK online seller that sells to individuals had a checkable price, and flammable liquids are awkward to post to the Isle of Man. Buy 99 % or stronger isopropyl alcohol locally (electronics, pharmacy or hardware shops), never 70 % rubbing alcohol.* Not verified: check the listing before you buy.	`local purchase`	1	not checked	a local shop
NE555P timer, DIP-8 The 1 kHz astable that drives the MOSFET gate in Lab 1.1.3. *Active part.*	`NE555P`	2	£0.44 + VAT	DigiKey UK

Item and use	SKU or part number	Qty	Price each	Seller
Computers, FPGA, debug and networking				
Raspberry Pi 5, 8 GB The lab host from Chapter 1.3 onwards; the Arm node in Units 2 and 4. *Memory shortages have raised the price; the 4 GB board (£105.60) is enough for Units 1 and 3 but not Unit 4.*	SC1112	1	£168.00 inc VAT	The Pi Hut
Raspberry Pi 27 W USB-C power supply, UK plug Powering the Pi 5. *Select the UK variant.*	SC1149	1	£11.50 inc VAT	The Pi Hut
Raspberry Pi Active Cooler for Pi 5 Keeping the Pi out of thermal throttling during kernel builds.	SC1148	1	£4.80 inc VAT	The Pi Hut
Raspberry Pi microSD card, 64 GB, A2, with Raspberry Pi OS Boot media for the Raspberry Pi 5 until it boots from NVMe in Chapter 2.7. *Substitution: no SanDisk or Samsung 64 GB A2 U3 card was found at a UK seller whose page could be fetched (The Pi Hut, Pimoroni and Maplin stock none; Amazon, Argos and Scan refused or did not render). The link opens the 64GB variant (SKU SC1629, in stock; the 32GB is sold out). A2 with SDR104 and command queueing, which the Pi 5 uses; the page does not state a U3 rating.*	SC1629	1	£30.60 inc VAT	The Pi Hut
Sensors				
Adafruit TMP117 ±0.1 °C temperature sensor (STEMMA QT) Drivers and metrology (1.3), firmware (2.6), device-tree overlays (2.7.5), Linux IIO (4.3). *TI TMP117. Few in stock: alternative SparkFun SEN-15805 at The Pi Hut, £16.30.*	ADA4821	2	£11.10 inc VAT	The Pi Hut
Adafruit HDC3022 temperature and humidity sensor (STEMMA QT) Humidity metrology and the salt test (1.3.2). *TI HDC3022 behind a PTFE filter. Sold out everywhere on 4 October 2026; DigiKey expected stock in February 2027. If it is still unavailable, buy the SHT45 below.*	ADA5989	1	£8.60 inc VAT	The Pi Hut
Adafruit Sensirion SHT45 temperature and humidity sensor (STEMMA QT) OPTIONAL The in-stock alternative to the HDC3022 for Lab 1.3.2. *Not a TI part. ±1.0 %RH, ±0.1 °C, I^2C at 0x44, CRC-8 on every word like the HDC3022.*	ADA5665	1	£12.60 inc VAT	Pimoroni
Adafruit INA228 85 V 20-bit power monitor Power metrology (1.3.3), the battery lab (1.4.4), the thrust stand (3.1.1), Linux hwmon (4.3). *TI INA228 with a 15 mΩ shunt on board. For the second board, desolder that shunt and use the Bourns 1 mΩ shunt.*	5832	2	£11.00 + VAT	DigiKey UK
SparkFun TMAG5273 3D Hall-effect sensor (Qwiic) Magnetic angle (1.3.4). *Uses the more sensitive TMAG5273A1 (±40 or ±80 mT) at I^2C 0x35.*	SEN-23880	1	£5.52 + VAT	DigiKey UK
Adafruit ADS1115 16-bit ADC (STEMMA QT) Sampling, aliasing and quantisation (1.2.3). *TI ADS1115.*	ADA1085	1	£14.40 inc VAT	The Pi Hut
Olimex BB-ADS1220 24-bit ADC breakout The load cell (1.2.4), the thrust stand (3.1.1), and the SPI IIO driver (4.3). *TI ADS1220. SPI mode 1 with a DRDY pin. Headers supplied but not soldered.*	BB-ADS1220	1	£13.57 + VAT	DigiKey UK
SparkFun 5 kg straight-bar load cell (TAL220B) The load-cell lab (1.2.4), which becomes the thrust stand (3.1.1). *Check the thread sizes in SparkFun's datasheet before making the mounting blocks.*	SEN-14729	1	£13.11 + VAT	DigiKey UK
Pololu OPT3101 three-channel time-of-flight module (no headers) Optical time of flight (1.3.5). *TI OPT3101. I^2C at 0x58, 2.5 to 5.5 V, up to 1 m on this module.*	3412	1	£34.28 + VAT	DigiKey UK
TI power				
TPS62933 3 A synchronous buck EVM Inside a buck converter (1.4.2). *5.1 to 30 V in, 5 V out at 3 A, 500 kHz. Switch-node and loop-injection test points.*	TPS62933EVM	1	£49.59 + VAT	DigiKey UK
TPS25947 eFuse EVM Inrush, short-circuit and reverse-current protection (1.4.3); the part Board A uses. *2.7 to 23 V, up to 5.5 A, jumper-set current limit.*	TPS25947EVM	1	£103.04 + VAT	DigiKey UK
LM74700-Q1 ideal-diode controller EVM Reverse-polarity protection (1.4.3). *Blocks −12 V and reacts within 2 µs.*	LM74700EVM	1	£51.96 + VAT	DigiKey UK
BQ4050 1–4S fuel gauge EVM OPTIONAL Stretch in 1.4: gauging the drone pack. *Needs the EV2400 and TI bqStudio on the Windows PC.*	BQ4050EVM-561	1	£82.72 + VAT	DigiKey UK

Item and use	SKU or part number	Qty	Price each	Seller
EV2400 SMBus/I²C gauge interface OPTIONAL Only with the BQ4050 stretch.	EV2400	1	£194.31 + VAT	DigiKey UK
RF test equipment and accessories				
tinySA Ultra spectrum analyser (ZS405) Spectrum and power measurement from Chapter 1.5 onwards; signal generator for 1.5.4. *Plus £7.50 postage. Mirfield is the authorised UK seller; clones are common elsewhere. Input limit +6 dBm absolute, 0 dBm in normal use.*	ZS405	1	£149.95 inc VAT	Mirfield Electronics
LiteVNA 64 vector network analyser S-parameters, antennas and matching (1.5.2, 1.5.3). *50 kHz to 6.3 GHz, so it covers 868 MHz and 2.4 GHz. Ships with cables and an SMA calibration kit. Never connect a transmitter to it.*	LiteVNA 64	1	€144.59 + import VAT	Eleshop (Netherlands)
RTL-SDR Blog V4 receiver The FM receiver (1.5.5), chirp measurement (3.3.3), packet demodulation (3.2.6). *Dongle only, 500 kHz to 1.766 GHz. Technofix is the authorised UK reseller.*	RTL-SDR-V4	1	£46.49 inc VAT	Technofix
Mini-Circuits 10 dB attenuator, 2 W, SMA, DC–6 GHz Protecting instruments; building attenuation chains.	VAT-10W2+	1	£27.76 + VAT	DigiKey UK
Mini-Circuits 20 dB attenuator, 2 W, SMA, DC–6 GHz As above.	VAT-20W2+	1	£33.13 + VAT	DigiKey UK
Mini-Circuits 30 dB attenuator, 2 W, SMA, DC–6 GHz The attenuator that always sits on a transmitter's output.	VAT-30W2+	1	£27.76 + VAT	DigiKey UK
Amphenol SMA 50 Ω termination, to 18 GHz VNA calibration checks and unused synthesiser outputs. *For a transmitter, fit the 30 dB attenuator first and the terminator after it.*	132360	3	£3.53 + VAT	DigiKey UK
Amphenol SMA to SMA cable, RG316, 12 in Connecting instruments, attenuators and boards. *Rated to 3 GHz, enough for 868 MHz and 2.4 GHz. The analysers include short cables too.*	135101-01-12.00	2	£14.70 + VAT	DigiKey UK
Amphenol SMA plug to SMA plug adapter Joining two female ports, such as attenuator to instrument.	132168	2	£7.03 + VAT	DigiKey UK
Linx 868 MHz half-wave whip antenna, SMA male VNA measurements (1.5.2) and the 868 MHz link (3.2). *853–883 MHz, 142 mm. Fits the CC1312R LaunchPad's SMA connector after the rework in Lab 3.2.1.*	ANT-868-CW-HWR-SMA	2	£9.90 + VAT	DigiKey UK
10 m RG-58 50 Ω coax, BNC plug to BNC plug The transmission line in the cable-reflections lab, Chapter 1.5. *Active; 951 in stock. How the four BNC pieces fit: at the near end the Pomona 2630 lead's BNC jack takes the cable's plug and its clips go to the driver and ground; at the far end the cable goes into one jack of the tee, the terminator into the other, and the tee's plug can go to scope channel 2. Remove the terminator to see an open end.*	115101-19-M10.0	1	£35.28 + VAT	DigiKey UK
BNC T adapter, plug to two jacks, 50 Ω Far-end junction in the cable-reflections lab, Chapter 1.5. *Active; 432 in stock. The link is DigiKey's part-number search, which lands on this product's page; the page did not expose its numeric product link.*	31-208-RFX	1	£5.97 + VAT	DigiKey UK
50 Ω BNC end terminator, plug Matched far-end load in the cable-reflections lab, Chapter 1.5. *Active; 3,864 in stock. Its plug fits the tee's spare jack, not the cable's plug. A feed-through terminator (jack to plug) such as Pomona 4119-50 would also work but costs £52.30.*	202103	1	£2.95 + VAT	DigiKey UK
BNC jack to two insulated crocodile clips, 127 mm Connects the SN74HC14N driver and its 47 Ω resistor to the near end of the coax in Chapter 1.5. *Active; 331 in stock. The BNC end is a jack (female) so it mates directly with the cable's plug; male-ended croc leads would need an extra adapter. The short 5-inch leads keep stray inductance out of the reflection measurement.*	2630	1	£15.15 + VAT	DigiKey UK
SMA jack (female) bulkhead, front-nut panel mount, solder cup Feed point of the quarter-wave antenna built on copper-clad board in Chapter 1.5. *Active; 2,876 in stock; £4.14 each at 10. 50 Ω, 18 GHz, nut included. Drill about 6.5 mm for its 1/4-36 thread, and solder the body to the copper round the hole so the ground plane is bonded.*	142-0701-401	2	£4.86 + VAT	DigiKey UK
Copper-clad FR4 board, double-sided, 152 × 152 mm, 1.6 mm Ground plane for the antenna lab in Chapter 1.5, and the low-pass-filter stretch. *Active; 156 in stock; £10.51 each at 10. 1 oz copper both sides.*	550	2	£12.79 + VAT	DigiKey UK

Item and use	SKU or part number	Qty	Price each	Seller
Solid-core hook-up wire, 22 AWG, 7.5 m spool (substitute for 1 mm copper wire) The quarter-wave radiator soldered to the SMA bulkhead in Chapter 1.5. *Substitution: no small reel of 1 mm (18 AWG) copper wire was found. DigiKey UK's smallest 18 AWG reels cost over £70 (CNC Tech 600218 magnet wire, 136.6 m, £102.03, alt_url; Alpha Wire 296 SV005 bus wire, 30.5 m, £73.53), and The Pi Hut, Pimoroni and Kunkune stock nothing near 1 mm. This 22 AWG (0.64 mm) solid wire, stripped, resonates almost the same; it is just less stiff, so keep the radiator straight. In stock; The Pi Hut's own brand.*	`104281`	1	£2.80 inc VAT	The Pi Hut
Drone airframe, power and flight electronics				
Overlander Supersport Pro 5000 mAh 3S 35C LiPo, XT60 Characterised in Lab 1.4.4; powers the drone in Chapter 3.5. *11.1 V, 343 g. Choose the XT60 option. Confirm the courier delivers lithium batteries to your address before ordering.*	`2577 (XT60)`	1	£45.98 inc VAT	Overlander
SkyRC S65 65 W AC balance charger Charging and storage-charging the pack. *Built-in mains supply, 2–4S, up to 6 A. Charge the 5000 mAh pack at 5 A (1C).*	`S65`	1	£32.05 inc VAT	HobbyRC
ToolkitRC LiPo safety bag, large Charging and storage. *Use it on a non-flammable surface. No bag fully contains a LiPo fire.*	`LiPo Safety Bag (Large)`	1	£3.90 inc VAT	HobbyRC
1–8S cell checker with low-voltage alarm Per-cell voltages before and after every use. *Plugs into the balance lead.*	`1-8S Cell Checker`	1	£2.60 inc VAT	HobbyRC
XT60 male connectors, pack of 2 Battery leads for the electronic load (1.4.4), the thrust stand and the drone.	`Black XT60 Male Connectors (2x)`	1	£1.50 inc VAT	HobbyRC
XT60 female connectors, pack of 2 As above.	`XT60 Female (2x)`	1	£1.50 inc VAT	HobbyRC
14 AWG silicone wire, red and black, 0.75 m each High-current leads: the battery lab, the thrust stand and the drone.	`14 AWG Flexible Silicone Wire`	2	£2.15 inc VAT	HobbyRC

Unit 2 kit

	Required	Optional
UK sellers, VAT included (ex-VAT prices × 1.2)	£420	£0
Overseas sellers, before import VAT and delivery	US$275	none
Items with no checked price	1	none

Not priced: Used rack server: Dell PowerEdge R730 (or R630, or Supermicro X11).

Item and use	SKU or part number	Qty	Price each	Seller
Components and consumables				
SN74HC00N quad NAND, DIP-14 The CMOS NAND comparison and the latch in Chapter 2.1. *Active; thousands in stock when checked.*	`SN74HC00N`	2	£0.57 + VAT	DigiKey UK
SN74HC08N quad AND, DIP-14 The adder in Lab 2.1.2. *Active; thousands in stock when checked.*	`SN74HC08N`	2	£0.58 + VAT	DigiKey UK
SN74HC32N quad OR, DIP-14 The adder and the state machine in Chapter 2.1. *Active; thousands in stock when checked.*	`SN74HC32N`	2	£0.54 + VAT	DigiKey UK
SN74HC74N dual D flip-flop, DIP-14 The setup-time violation and the state machine in Chapter 2.1. *Active; thousands in stock when checked.*	`SN74HC74N`	2	£0.83 + VAT	DigiKey UK
SN74HC86N quad XOR, DIP-14 The adder in Lab 2.1.2. *Active; thousands in stock when checked.*	`SN74HC86N`	2	£0.83 + VAT	DigiKey UK
BS250P 45 V P-channel MOSFET, TO-92 The P-channel half of the discrete CMOS inverter and NAND in Chapter 2.1. *Status Last Time Buy (DigiKey shows a last-buy date of 06/01/2027) but orderable: 20,051 in stock; £0.816 each at 10. Threshold up to −3.5 V, 14 Ω at a −10 V gate. If it has gone, the active replacement with the same 3.5 V maximum threshold is Microchip VP2106N3-G (TO-92, 60 V, 12 Ω), £0.47, 4,327 in stock TP0610L is not listed at DigiKey UK. Check its pin order against the 2N7000's before building the inverter.*	`BS250P`	5	£1.29 + VAT	DigiKey UK

Item and use	SKU or part number	Qty	Price each	Seller
8-position DIP slide switch, through-hole, 2.54 mm Adder inputs and latch tests in Chapter 2.1. *Active; 7,139 in stock. Standard DIP body that straddles the breadboard's centre channel like a chip.*	`206-8ST`	2	£1.15 + VAT	DigiKey UK
10 kΩ single-turn cermet trimmer, top adjust, through-hole Input sweep for the discrete CMOS inverter and the adjustable delay in the setup-time lab, Chapter 2.1. *Active; 12,049 in stock; sold singly; £1.12 each at 10. Turn it with a small flat screwdriver. The cheaper 3362P-1-103LF is sold only in tubes of 50. Alternative with a finger knob: The Pi Hut breadboard-friendly trim potentiometer, 10K variant (alt_url, SKU 102831), £0.80 inc VAT, in stock.*	`3386P-1-103LF`	3	£1.34 + VAT	DigiKey UK
Computers, FPGA, debug and networking				
Radiona ULX3S FPGA board, ECP5 85F Chapters 2.2 to 2.4: HDL, your CPU, caches and SDRAM. *Plus about $18 shipping. 84k LUTs, 32 MB SDRAM, FT231X USB-JTAG and serial.*	`ULX3S 85F`	1	US$275.00 + import VAT	Crowd Supply
Adafruit EMC2101 fan controller and temperature sensor (STEMMA QT) Fan control from firmware (2.6) and from a Linux driver (4.3). *Microchip EMC2101: PWM output, tachometer input, internal and external-diode temperature.*	`ADA4808`	1	£5.40 inc VAT	Pimoroni
Raspberry Pi M.2 HAT+ NVMe on the Pi 5 for the boot chain (2.7) and the second storage node (4.5). *PCIe 2.0 ×1, takes 2230 and 2242 drives.*	`SC1166`	1	£11.50 inc VAT	The Pi Hut
Raspberry Pi SSD 256 GB, M.2 2230 The Pi's boot and data drive. *Out of stock at The Pi Hut, Pimoroni and Rapid in October 2026. Any 2230 or 2242 NVMe drive fits; the in-stock fallback is the Kingston NV3 1 TB 2230 at Scan, £219.98.*	`SC1439`	1	£57.60 inc VAT	The Pi Hut
LetsTrust TPM 2.0 for Raspberry Pi Measured boot and attestation on the Pi (2.7.4, 2.7.5) and in the capstone. *Now built with the Infineon SLB 9672 (the successor to the SLB 9670). Pi 5 support is not stated by the seller; check before relying on it, and check clearance over the M.2 HAT+.*	`The Pi Hut 104062`	1	£25.90 inc VAT	The Pi Hut
SparkFun ATECC608A cryptographic co-processor (Qwiic) Device identity for the BMC firmware (2.6, 3.4, 4.7). *Locking its configuration is permanent: experiment on one, keep the other.*	`DEV-18077`	2	£3.72 + VAT	DigiKey UK
Raspberry Pi Debug Probe (CMSIS-DAP, SWD and UART) The Pi 5's debug UART (2.7.5); SWD for your own boards (3.4). *Works with OpenOCD's cmsis-dap interface.*	`SC0889`	1	£11.50 inc VAT	The Pi Hut
Waveshare industrial USB-to-TTL converter, FT232RL Serial consoles for your boards and for logging the drone's UARTs. *Set the switch to 3.3 V before connecting.*	`WAV-17939`	2	£6.80 inc VAT	The Pi Hut
TI microcontroller boards				
MSP432E401Y LaunchPad (Cortex-M4F, Ethernet) Bare-metal firmware (2.6); the drone's flight controller (3.5). One stays on the bench. *120 MHz, 1 MB flash, 256 KB SRAM, 10/100 Ethernet MAC and PHY, AES and SHA accelerators, XDS110 debugger with a UART back-channel.*	`MSP-EXP432E401Y`	2	£40.04 + VAT	DigiKey UK
MSPM0G3507 LaunchPad (Cortex-M0+) The second-architecture lab (2.6.7). *OpenOCD supports it in git master only. DigiKey showed back-order; order early.*	`LP-MSPM0G3507`	1	£17.19 + VAT	DigiKey UK
Server, storage and networking				
Used rack server: Dell PowerEdge R730 (or R630, or Supermicro X11) x86 bare metal (2.5.3), Secure Boot and measured boot (2.7), kernels, virtualisation and data paths (Unit 4). *Buy using the Handbook's server checklist: v4 Xeons, 64 GB or more registered ECC, iDRAC8 Enterprise, an HBA330 or H730 in HBA mode, a TPM header, two power supplies. A 2U R730 is quieter than a 1U R630 and takes full-height cards. No price could be checked.* Not verified: check the listing before you buy.	`PowerEdge R730, Xeon E5-26xx v4, iDRAC8 Enterprise`	1	not checked	eBay UK (choose a UK refurbisher that accepts returns)
Intel X520-DA2 dual-port SFP+ 10 GbE adapter (new) PCIe by hand (2.4.4); SR-IOV (4.4); the network data path (4.5). *Intel 82599ES. New from Scan avoids the counterfeits common on auction sites. Both brackets included.*	`E10G42BTDABLK`	1	£69.98 inc VAT	Scan
Ubiquiti SFP+ passive direct-attach cable, 1 m Looping the X520's two ports together. *Pre-order at Scan; the StarTech 1 m cable (£33.49) ships sooner.*	`UACC-DAC-SFP10-1M`	1	£16.49 inc VAT	Scan
TP-Link TL-SG108E 8-port managed gigabit switch Lab VLANs for netboot (2.7.6) and the capstone's three networks (4.7). *802.1Q VLANs and port mirroring.*	`TL-SG108E`	1	£28.00 inc VAT	Scan

Item and use	SKU or part number	Qty	Price each	Seller
Cat6 UTP patch cable, RJ45, 1 m The lab network from Chapter 2.7: server, Pi, switch, Boards A and B. *Available when checked. Maplin's own brand; the sku is the variant's EAN, and the link opens the 1 m variant. The listing does not say whether the conductors are solid copper or copper-clad aluminium. DigiKey UK's 1 m Cat6 cables were £19 or out of stock, and The Pi Hut sells Cat6A only in 2 m (£3).*	5055189044449	5	£3.99 inc VAT	Maplin
Cat6 UTP patch cable, RJ45, 3 m, black Longer runs from the bench to the switch and your home router. *Available when checked. A different Maplin listing from the 1 m cable, because that one has no 3 m length; the sku is the variant's EAN and the link opens the 3 m variant. 1 Gb/s, 250 MHz UTP; conductor material not stated.*	5026686195712	2	£5.49 inc VAT	Maplin

Unit 3 kit

	Required	Optional
UK sellers, VAT included (ex-VAT prices × 1.2)	£1,648	£154
Overseas sellers, before import VAT and delivery	US$199	US$125
Items with no checked price	1	none

Not priced: Compact USB power bank, about 5,000 mAh, no longer than 105 mm, bought locally.

Item and use	SKU or part number	Qty	Price each	Seller
Bench instruments and tools				
Andonstar AD246S-M HDMI digital microscope with stand, UK plug Inspecting joints and 0402 parts: the SMA rework (3.2.1) and board bring-up (3.4). *Use lens A (18–720×, 12–320 mm working distance).*	AD246S-M	1	US$199.00 + import VAT	Andonstar
SkyRC TLD001 two-channel K-type temperature logger OPTIONAL Logging motor and ESC temperatures on the thrust stand (3.1). *Two probes, −20 to 210 °C, Bluetooth logging to a phone.*	TLD001	1	£19.99 inc VAT	Flying Tech
Hot-air rework station, 700 W, UK plug OPTIONAL A cheaper hot-air station if you bought the Pinecil instead of the ATTEN ST-8802. *6 in stock, reduced from £49. UK plug, 220–240 V, 100–500 °C, up to 120 L/min, brushless fan in the handle; 5, 8 and 10 mm nozzles; 12-month guarantee; ships from the UK. Kunkune shows its VAT number but the price is not labelled; UK consumer prices include VAT by law, so treat £39 as VAT-inclusive. The Quick 861DW could not be verified at any UK seller.*	858D	1	£39.00 inc VAT	Kunkune
Computers, FPGA, debug and networking				
Dell OptiPlex 3070 Micro, i5-9400T, 8 GB, 256 GB NVMe, refurbished, Windows 11 Pro TI's Windows-only tools: SmartRF Studio 7, TICS Pro, UniFlash for mmWave, mmWave Studio, bqStudio. *Choose Windows 11 Pro in the operating-system option. Needs a monitor, keyboard and mouse. Any x86-64 Windows 10 or 11 PC you already own does the same job; Windows on an Apple-silicon Mac does not, because the debug-probe drivers are x86 only.*	Tier1Online 292-3070-MICRO-i59T	1	£189.00 inc VAT	Tier1Online
Raspberry Pi Compute Module 5, 4 GB RAM, 32 GB eMMC, wireless The computer on Board B (3.4.5, 3.4.7) and the Arm node in the capstone. *Stock was not shown for this variant. Any CM5 with eMMC works; buy it only once Board B's design has passed review.*	SC1595	1	£134.40 inc VAT	The Pi Hut
TI microcontroller boards				
F280025C LaunchPad (C2000 real-time MCU) Your own field-oriented motor control (3.1.2, 3.1.3). *100 MHz C28x with FPU and trigonometric unit. Needs Code Composer Studio and TI's C2000 compiler; char is 16 bits.*	LAUNCHXL-F280025C	1	£28.51 + VAT	DigiKey UK
TI motor control				
DRV8323RS three-phase inverter BoosterPack The power stage for your FOC firmware, on the bench motor and then a drone motor. *6 to 54 V, 15 A continuous, 20 A peak, three current-sense amplifiers, SPI-configured gate driver.*	BOOSTXL-DRV8323RS	1	£81.92 + VAT	DigiKey UK
24 V sensorless BLDC lab motor A known motor for bringing up FOC before you try a drone motor. *NEMA 17, 24 V, four pole pairs, no Hall sensors or encoder, 3 A maximum. DigiKey had three in stock.*	LVBLDCMTR	1	£49.59 + VAT	DigiKey UK

Item and use	SKU or part number	Qty	Price each	Seller
TI radio, microwave and radar				
CC1312R LaunchPad, 868/915 MHz Both ends of the 868 MHz telemetry link (3.2, 3.5). *Up to +14 dBm, −110.5 dBm sensitivity at 50 kbps. RF goes to the PCB antenna; moving C36 to C37 routes it to SMA connector J7 (TI SWRA588A section 5.3). XDS110 debugger on board.*	`LAUNCHXL-CC1312R1`	2	£36.79 + VAT	DigiKey UK
LMX2572 6.4 GHz low-power synthesiser EVM A PLL you program, its spurs and lock time, and a chirp you can measure (3.3). *12.5 MHz to 6.4 GHz with FSK and ramp generation. Kit includes the Reference PRO board (100 MHz reference and USB programming for TICS Pro), two SMA adaptors, USB cable and ribbon. Needs 3.3 V at up to 150 mA into its VCC SMA.*	`LMX2572EVM`	1	£288.84 + VAT	DigiKey UK
IWRL6432 low-power 60 GHz radar BoosterPack EVM FMCW radar (3.3.5, 3.3.6) and the drone's radar altimeter (3.5.7). *57–64 GHz, 2 transmit and 3 receive antennas, XDS110 on board, powered from USB. Uses MMWAVE-L-SDK.*	`IWRL6432BOOST`	1	£130.13 + VAT	DigiKey UK
FTDI USB to SPI/I²C/JTAG cable, 3.3 V (FT232H) Streaming the radar's raw ADC samples over SPI (3.3.6). *The cable TI's MMWAVE-L-SDK names for SPI ADC streaming.*	`C232HM-DDHSL-0`	1	£26.13 + VAT	DigiKey UK
RF test equipment and accessories				
Mini-Circuits 30 dB attenuator, two more Reaching receiver sensitivity in the packet-error test (3.2.3): up to 120 dB of fixed attenuation with the others.	`VAT-30W2+`	2	£27.76 + VAT	DigiKey UK
Pasternack SMA male to double banana cable, 3 ft OPTIONAL Feeding 3.3 V from the bench supply into the LMX2572EVM's VCC SMA. *A spare SMA cable with one end cut and stripped does the same job for nothing.*	`PE3W01980-36`	1	£54.98 + VAT	DigiKey UK
SMA female-to-female straight adapter (jack to jack), 50 Ω Joining two male-ended SMA cables or attenuators in the Unit 3 RF set-ups. *Active; 11,882 in stock; £4.99 each at 10. DigiKey's attribute table lists both ends as 'SMA Jack, Female Socket', straight, 18 GHz. Amphenol RF's datasheet link returned HTTP 403, so the gender is confirmed from DigiKey's listing rather than the datasheet itself. Do not confuse it with 132168, which is male to male.*	`132169`	2	£5.87 + VAT	DigiKey UK
SMA male to SMA male cable, RG-316, 305 mm (12 in) Extra test-port cables for the LiteVNA and tinySA in Unit 3. *Active; 6,460 in stock. Standard-polarity SMA plug both ends, 50 Ω, rated to 8 GHz. Cheaper than the Amphenol 135101-01-12.00, which DigiKey UK still shows at £14.70 with 1,168 in stock.*	`CSE-SGAM-305-SGAM`	2	£6.76 + VAT	DigiKey UK
Die-cast aluminium box, 119.5 × 94.0 × 56.5 mm, unpainted RF isolation box for a LaunchPad running from a power bank in Unit 3. *Active; 264 in stock plus 1,214 at the factory. Dimensions are outside; the inside is smaller by the wall thickness, so check that the LaunchPad and power bank fit stacked. The shallower 1590BB (119.6 × 94.0 × 41.9 mm) is £11.03.*	`1590C`	1	£12.95 + VAT	DigiKey UK
Compact USB power bank, about 5,000 mAh, no longer than 105 mm, bought locally Powers the transmitting LaunchPad inside the sealed box in Lab 3.2.3. *Online UK sellers checked would not post lithium power banks to the Isle of Man, and the common 10,000 mAh models are too long for the 1590C box. Buy one locally: no longer than about 105 mm, with a USB-A or USB-C output and, ideally, a low-current mode so it does not switch off under the LaunchPad's small load.* Not verified: check the listing before you buy.	`local purchase`	1	not checked	a local shop
Drone airframe, power and flight electronics				
Tarot FY450 450 mm quadcopter frame with integrated power board The airframe (3.5) and the one-axis rig (3.1.7). *282 g, glass-fibre plates, 16/19 mm M3 motor holes, takes 10-inch props.*	`FY450`	1	£20.00 inc VAT	Flying Tech
SunnySky X2212-13 980KV II brushless motor Thrust stand and FOC tests (3.1), the drone (3.5). *58 g, 15 A for 30 s. On 3S with a 10-inch prop: 300 g of thrust at 3.0 A, 910 g at 15.5 A (maker's table). Do not use 4S with 10-inch props: the current exceeds the rating. Check the 16 × 19 mm mounting on arrival.*	`X2212-13 KV980 II`	4	£19.60 inc VAT	Flying Tech
EPP 10 × 4.5 propellers, CW and CCW pair Two pairs fly; two pairs are spares. *Use the 5 mm adapter ring on the SunnySky prop adapter.*	`EPP1045`	4	£4.20 inc VAT	Flying Tech
Little Bee CloudPhoenix 35 A BLHeli_32 ESC, 3–6S The flying ESCs, driven by your DShot firmware (3.5.1). *35 A continuous, DShot up to 1200, telemetry. No 5 V output: the electronics need their own supply.*	`CloudPhoenix 35A`	4	£19.95 inc VAT	HobbyRC
Matek Micro BEC 6–30 V to 5 V, 1.5 A, pack of 3 5 V for the flight electronics until your Board P flies.	`Micro BEC 6-30V`	1	£6.75 inc VAT	HobbyRC

Item and use	SKU or part number	Qty	Price each	Seller
RadioMaster Pocket transmitter, ExpressLRS 2.4 GHz The pilot's control link (3.5.2). *Needs two 18650 cells.*	`Pocket ELRS`	1	£71.49 inc VAT	HobbyRC
RadioMaster 3200 mAh 18650 cells, pair Powering the transmitter. *Charge inside the transmitter. Unprotected cells only: protected ones do not fit.*	`18650 3200mAh x2`	1	£11.99 inc VAT	HobbyRC
RadioMaster RP1 V2 ExpressLRS receiver, CRSF output The receiver whose CRSF stream your firmware parses (3.5.2). *Update it over Wi-Fi to the transmitter's ExpressLRS major version before binding.*	`RP1 V2`	1	£18.99 inc VAT	HobbyRC
Adafruit LSM6DSOX 6-DoF IMU (SPI or I²C) Inertial sensing (3.1.4, 3.1.5) and the flight controller until Board F carries the chip. *ST LSM6DSOX: gyro to ±2000 dps, accelerometer to ±16 g, up to 6.7 kHz. TI makes no MEMS IMU.*	`ADA4438`	2	£11.40 inc VAT	Pimoroni
SparkFun BMP581 pressure sensor (Qwiic) Altitude fusion with the radar (3.5.7). *Bosch BMP581, ±0.5 hPa. Power from 3.3 V only. Cover it with open-cell foam against prop wash.*	`SEN-20170`	1	£20.10 inc VAT	The Pi Hut
Adafruit LIS3MDL triple-axis magnetometer Heading (3.1.6), mounted on a mast away from the power wiring. *ST LIS3MDL, magnetoresistive, ±4 to ±16 gauss.*	`ADA4479`	1	£9.60 inc VAT	The Pi Hut
Matek M10Q-5883 GNSS and compass module OPTIONAL Stretch in 3.5: position logging and position hold. *u-blox M10 on a UART, QMC5883L compass on I²C.*	`M10Q-5883`	1	£29.10 inc VAT	HobbyRC
AmovLab drone PID tuning rig, up to 450 mm OPTIONAL Three-axis rig tuning (3.5.5). The one-axis rig you build in 3.1.7 is enough. *No load limit published: ask the seller whether it takes 1.2 kg before ordering.*	`PID Test Rig`	1	US$125.00 + import VAT	AmovLab
Adafruit nylon M3 screw and standoff set, 420 pieces Mounting LaunchPads and boards on the frame and the rig.	`ADA4685`	1	£16.30 inc VAT	The Pi Hut
Cable ties, 100 × 2.5 mm, pack of 100 Wiring on the frame.	`103060`	1	£1.50 inc VAT	The Pi Hut
Double-sided foam tape, 12 mm × 10 m Vibration isolation for the flight controller; mounting the receiver. *Choose the 12 mm variant.*	`103426`	1	£2.50 inc VAT	The Pi Hut
JST-GH 4-pin to 4-pin cable, 200 mm I2C and CAN links between the flight controller and peripherals in Unit 3. *124 in stock; sold singly (£1.67 ex VAT). No manufacturer part number published, so the sku is the listing title. Length is 200 mm, not 150 mm: no 150 mm assembled JST-GH cable was in stock at a UK hobby seller. HobbyRC lists only JST-SH, and Unmanned Tech's 5-pack of 200 mm cables (£5.49, alt_url) is on backorder. Check with the meter that pin 1 goes to pin 1 before joining two ports.*	`Matek Systems JST-GH-4Pin to 4Pin 20cm Cable`	5	£2.00 inc VAT	3DXR
JST-GH 6-pin to 6-pin cable, 200 mm UART (telemetry and GPS) links on the flight controller in Unit 3. *85 in stock; sold singly (£1.83 ex VAT). No manufacturer part number published, so the sku is the listing title. 200 mm for the same reason as the 4-pin cable; Unmanned Tech's 5-pack is on backorder. Check pin 1 to pin 1 with the meter before use.*	`Matek Systems JST-GH-6Pin to 6Pin 20cm Cable`	5	£2.20 inc VAT	3DXR

Unit 4 kit

	Required	Optional
UK sellers, VAT included (ex-VAT prices × 1.2)	£173	£0
Overseas sellers, before import VAT and delivery	none	none
Items with no checked price	1	none

Not priced: Intel SSD DC P4510 1 TB U.2, used.

Item and use	SKU or part number	Qty	Price each	Seller
Sensors				
Adafruit OPT4048 XYZ colour and lux sensor (STEMMA QT) The IIO driver you write in Chapter 4.3 (and the Unit 1 stretch). *TI OPT4048. Linux has no driver for it. Alternative: SparkFun SEN-22638, £13.00.*	`ADA6335`	1	£9.30 inc VAT	The Pi Hut

Item and use	SKU or part number	Qty	Price each	Seller
Server, storage and networking				
Kingston NV3 500 GB M.2 2280 NVMe (to be sacrificed) Power-loss and write-amplification experiments in Chapter 4.5. *Consumer drive with no power-loss protection, which is the point.*	`SNV3S/500G`	1	£88.99 inc VAT	Scan
ICY DOCK EZConvert Ex Pro M.2 NVMe to PCIe ×4 adapter Fitting the M.2 drive in the server, which has no M.2 slot.	`MB987M2P-1B`	1	£35.99 inc VAT	Scan
Intel SSD DC P4510 1 TB U.2, used The drive with power-loss protection for the storage data path (4.5) and the capstone. *Ask the seller for a SMART report showing percentage used, power-on hours and media errors. No price could be checked.* Not verified: check the listing before you buy.	`SSDPE2KX010T8`	1	not checked	eBay UK
StarTech U.2 to PCIe 3.0 ×4 adapter Fitting the U.2 drive in the server. *Full-height bracket only: needs a full-height slot (R730 or a 2U Supermicro).*	`PEX4SFF8639`	1	£38.99 inc VAT	Scan

Fabrication budget

Custom boards have no part number until you design them, so the course budgets for them instead. These are estimates for five boards of each design at a low-cost fabricator in late 2026, before delivery. Get real quotes from your fabricator's website with your Gerber files, and log them for Chapter 4.6.

Board	What is ordered	Estimate
P	Two layers, 2 oz copper, bare boards; you solder the parts yourself	£15 to £30, plus parts about £25 a board
F	Four layers, assembly of the IMU, barometer, flash and regulator; connectors hand-soldered	£60 to £120
A	Four layers, assembly of the MCU, Ethernet magnetics and fine-pitch parts	£90 to £180
B	Six layers with controlled impedance, assembly of the CM5 connectors and the M.2 slot	£180 to £350

Buying a used rack server

The server is bought second-hand. The recommendation is a Dell PowerEdge R730 (2U), which is quieter than the 1U R630 and has full-height PCIe slots for the U.2 adapter. A Supermicro X11 board in a 2U chassis is the alternative. Search a UK refurbisher or eBay UK with terms such as `dell r730 e5-2680 v4 idrac8 enterprise`, filter to UK sellers that accept returns, and check sold listings to see what they actually sell for.

BEFORE YOU BUY

Every line must be true of the listing, or confirmed by the seller in writing.

- ☐ The model and generation are clear from photographs or the service tag. Avoid the R620 and R720, which are an older generation with iDRAC7 and DDR3.
- ☐ The CPUs are Xeon E5-2600 v4 (Dell) or Xeon Scalable (Supermicro X11). Both support VT-x, VT-d and SR-IOV. Both sockets are populated if you want every memory slot usable, with both heatsinks.
- ☐ At least 64 GB of registered ECC DDR4, with no mixed module types, shown in a photograph or a screenshot.
- ☐ The BMC licence is stated: iDRAC8 Enterprise on Dell, which you need for the remote console. Ask for the iDRAC or BMC firmware version; recent versions serve Redfish.
- ☐ A storage controller that can pass drives straight through (HBA330, H330, or H730 in HBA mode), with drive caddies.
- ☐ PCIe risers fitted, with a full-height slot for the U.2 adapter and a slot for the X520.
- ☐ Two power supplies and UK C13 mains leads.

- [ ] A TPM fitted, or a TPM header on the board. Dell 13th-generation TPM modules bind to the board they are fitted to, so buy one with the server or a new sealed module.
- [ ] Photographs of the iDRAC or BMC system summary with no hardware errors.
- [ ] Rails if you will rack it, and a delivery quote for 20 to 30 kg to your address.

When it arrives, enable virtualisation, VT-d and SR-IOV in the BIOS, update the BIOS and iDRAC firmware, and record the service tag and every component in your notebook.

Things you make

- **The load-cell and thrust stand (Labs 1.2.4 and 3.1.1).** Two blocks of 18 mm plywood to mount the bar, bolted with the screws the load cell's datasheet specifies, on a base heavy enough not to move: a 400 × 200 mm offcut of 18 mm plywood with the base block screwed down. The free end carries a 120 × 120 mm plywood platform.
- **The one-axis rig (Labs 3.1.7 and 3.5.5).** An A-frame of 18 mm plywood about 400 mm tall carrying a 10 mm steel rod through two pipe clips, with the rod passing through the frame's centre plates; mechanical stops at ±30° made from two blocks.
- **The RF box (Lab 3.2.3).** Drill the die-cast box's end for the SMA bulkhead (a 6.5 mm hole for a 1/4-36 thread), fit the bulkhead with its washer for a ground contact, and tape the lid seam with copper tape.
- **The magnetometer mast (Lab 3.1.6).** A 10 mm wooden dowel or carbon tube at the height your measurements choose, with the LIS3MDL at its top and its cable twisted down the mast.
- **The tether (Lab 3.5.6).** 1.5 m of paracord tied to the frame's centre plates and to a 10 kg anchor such as a kettlebell or a bucket of sand.

PART 5

Setup

Four computers do the work: your workstation (a Mac), the Raspberry Pi 5 on the bench, a small Windows PC for TI's Windows-only tools, and, from Chapter 2.4, the rack server. Set each up as it arrives. The checklist at the end of this part is the done-when for the whole handbook.

The workstation

macOS on Apple silicon runs almost everything: the toolchains, the FPGA flow, KiCad, Code Composer Studio, the simulators, Python and Typst. Kernel and hypervisor work happens only on the server, because the workstation holds your personal files. Install Homebrew, then:

```
brew install git cmake ninja python@3.13 pkg-config autoconf automake libtool \
  libusb hidapi qemu verilator icarus-verilog ngspice picocom tio typst
brew install --cask kicad gcc-arm-embedded
brew tap riscv-software-src/riscv && brew install riscv-tools
```

- **FPGA tools.** The distribution versions of Yosys and nextpnr lag behind, so use the OSS CAD Suite build for `darwin-arm64` from YosysHQ's releases, unpacked to `~/opt/oss-cad-suite` and added to your `PATH`. It includes Yosys, nextpnr-ecp5, openFPGALoader, Verilator and SymbiYosys.
- **OpenOCD from source.** The last OpenOCD release predates the MSPM0 support and the new names of TI's board files, so build git master:

  ```
  git clone https://github.com/openocd-org/openocd.git && cd openocd
  ./bootstrap && ./configure --enable-xds110 --enable-cmsis-dap && make -j8 && sudo make
  install
  ```

- **TI's Code Composer Studio** (version 21 or later, native on Apple silicon), with support for MSP432, C2000 and the SimpleLink CC13xx devices. Inside it, install C2000Ware with the Motor Control SDK (Chapter 3.1) and the SimpleLink Low Power F2 SDK (Chapter 3.2).
- **Logic analysis.** PulseView from the sigrok project's download page; optionally Saleae Logic 2.
- **Python.** A virtual environment for the course's scripts:

  ```
  python3 -m venv ~/venvs/omh && source ~/venvs/omh/bin/activate
  pip install numpy scipy matplotlib pandas pyserial pyftdi pyyaml jupyterlab
  ```

- **Simulators in the browser or on the desktop.** Falstad (Chapter 1.1), Digital by hneemann (Chapter 2.1), Surfer or GTKWave for waveforms.

The Raspberry Pi 5

Install the current 64-bit Raspberry Pi OS with Raspberry Pi Imager, with SSH enabled and your key installed. Then enable I²C and SPI in `raspi-config`, fit the Active Cooler, and install:

```
sudo apt install -y git build-essential python3-venv python3-dev i2c-tools \
  gpiod libgpiod-dev device-tree-compiler minicom
python3 -m venv ~/venv && ~/venv/bin/pip install smbus2 spidev gpiod numpy scipy matplotlib
pyserial
```

Plug the TMP117 into the Qwiic cable on the Pi's I²C pins and run `i2cdetect -y 1`; it should appear at 0x48. From Chapter 2.7 the Pi also boots from NVMe through the M.2 HAT+, and from Chapter 4.2 you build its kernel.

The Windows PC

Several TI tools run only on x86 Windows: SmartRF Studio 7 for the CC1312R (Chapter 3.2), TICS Pro for the LMX2572 (Chapter 3.3), the mmWave visualiser for the IWRL6432 (Chapter 3.3), and bqStudio for the optional fuel-gauge stretch. UniFlash, which flashes the IWRL6432, also runs on macOS and Linux, but it is simplest to install it on the same PC. Windows on an Apple-silicon Mac does not work for them, because the debug-probe drivers are x86 only. Set up the refurbished OptiPlex (or any x86 Windows 10 or 11 PC) with:

- SmartRF Studio 7, TICS Pro and UniFlash from TI's website, which also installs the XDS110 drivers;
- MMWAVE-L-SDK for the IWRL6432 and its visualiser;
- FTDI's D2XX driver for the C232HM cable (Chapter 3.3);
- Python, so the same scripts run on it when needed.

Keep it on the bench network. It needs no other software.

The server

When the server arrives (Chapter 2.4 needs it), work through this list before installing anything else.

1. In the BIOS: enable virtualisation (VT-x), VT-d, SR-IOV and the TPM; set the boot mode to UEFI.
2. Update the BIOS and the iDRAC (or Supermicro BMC) firmware to the latest release.
3. Give the iDRAC a static address on the management VLAN, enable Redfish and serial-over-LAN, and change the default password.
4. Install Debian stable or Ubuntu Server LTS on a small drive, leaving the NVMe drives for the labs, then install the tools:

   ```
   sudo apt install -y build-essential git cmake ninja-build python3-venv \
     qemu-system qemu-utils ovmf swtpm swtpm-tools tpm2-tools \
     gnu-efi sbsigntool efitools mtools dosfstools xorriso \
     libelf-dev libssl-dev bc flex bison dwarves \
     trace-cmd bpftrace fio iperf3 nvme-cli pciutils \
     dnsmasq ipxe acpica-tools dmidecode ipmitool
   ```

   Add `perf` from your distribution's package: `linux-perf` on Debian, `linux-tools-generic` on Ubuntu.
5. Confirm `lscpu | grep vmx`, `dmesg | grep -i -e DMAR -e IOMMU`, and `/dev/tpm0`.

The lab network

The TL-SG108E carries four VLANs from Chapter 2.7 onwards. Assign them once and keep the plan in your notebook.

VLAN	Name	What lives on it
10	management	The server's iDRAC, Board A, the node agents
20	lab	Netboot and provisioning (dnsmasq and iPXE), your workstation
30	storage	NVMe over Fabrics and replication traffic (Chapter 4.5)
40	tenant	Virtual machines' traffic in the capstone (Chapter 4.7)

Port	Connection
1	Workstation: VLAN 20 untagged, 10 tagged
2	Server iDRAC: VLAN 10 untagged
3	Server Gigabit port: VLANs 20, 30 and 40 tagged
4	Raspberry Pi 5: VLANs 20, 30 and 40 tagged

Port	Connection
5	Board A: VLAN 10 untagged
6	Board B: VLANs 20, 30 and 40 tagged
7 and 8	Spare, or your home network uplink on VLAN 20 if you need internet access on the lab network

The labs repository

Create the repository with the layout in Part 2, push it to a private remote, and add a `setup/notebook.md` recording what arrived, what was substituted, and what is still on back-order. That is your first supply-chain log.

SETUP CHECKLIST

Setup is done when every line is true. Tick the server lines when it arrives.

- [ ] **Bench:** meter, oscilloscope, supply, generator and load power up and pass a self-test; the logic analyser shows a signal in PulseView; the soldering station heats and holds temperature.
- [ ] **Workstation:** `yosys -V`, `nextpnr-ecp5 --version`, `verilator --version` and `sby --help` run from the OSS CAD Suite.
- [ ] **Toolchains:** `arm-none-eabi-gcc --version` runs, and `riscv64-unknown-elf-gcc -march=rv32i -mabi=ilp32 -c hello.c` succeeds.
- [ ] **OpenOCD:** your git-master build connects to the MSP-EXP432E401Y with `openocd -f board/ti/msp432-launchpad.cfg` and reads the device ID.
- [ ] **Emulation:** `qemu-system-x86_64` with OVMF reaches the UEFI shell, and `qemu-system-riscv64 -M virt -nographic -bios default` prints the OpenSBI banner.
- [ ] **Code Composer Studio** builds and flashes a C2000Ware example to the F280025C LaunchPad.
- [ ] **Windows PC:** SmartRF Studio 7 detects a CC1312R LaunchPad over its XDS110, and TICS Pro opens the LMX2572 profile.
- [ ] **Pi:** boots with SSH, I²C and SPI enabled, and `i2cdetect -y 1` shows the TMP117 at 0x48.
- [ ] **FPGA:** `openFPGALoader -b ulx3s blink.bit` loads an example bitstream and an LED blinks.
- [ ] **Server:** iDRAC reachable on VLAN 10; serial-over-LAN works; Linux installed; `vmx`, the IOMMU and `/dev/tpm0` all present.
- [ ] **Repository:** the labs repository exists with the Part 2 layout, and the setup notebook holds the supply-chain log.

PART 6

Reference library

Each chapter's reading list names the sections to read and when. These are the books worth owning, and the free documents worth downloading before you start. A book appears once, against every chapter that uses it.

Books to buy

Book	Chapters
Horowitz and Hill, *The Art of Electronics*, 3rd ed.	1.1, 1.2, 1.4, 3.4
Fraden, *Handbook of Modern Sensors*, 5th ed.	1.3
Erickson and Maksimović, *Fundamentals of Power Electronics*, 3rd ed.	1.4 (for depth)
Bogatin, *Signal and Power Integrity, Simplified*, 3rd ed.	1.5, 3.4
Pozar, *Microwave Engineering*, 4th ed.	1.5
Lyons, *Understanding Digital Signal Processing*, 3rd ed.	1.5, 3.2, 3.3
Harris and Harris, *Digital Design and Computer Architecture*, RISC-V ed.	2.1, 2.2, 2.3
Patterson and Hennessy, *Computer Organization and Design*, RISC-V ed.	2.3, 2.4
Hennessy and Patterson, *Computer Architecture: A Quantitative Approach*	2.3, 2.4
Jackson and Budruk, *PCI Express Technology 3.0* (MindShare)	2.4, 4.3
White, *Making Embedded Systems*, 2nd ed.	2.6
Yiu, *The Definitive Guide to Arm Cortex-M3 and Cortex-M4 Processors*	2.6
Zimmer, Rothman and Marisetty, *Beyond BIOS*, 3rd ed.	2.7
Hanselman, *Brushless Motors: Magnetics, Design and Control*	3.1
Franklin, Powell and Emami-Naeini, *Feedback Control of Dynamic Systems*	3.1 (or the free Åström and Murray)
Beard and McLain, *Small Unmanned Aircraft: Theory and Practice*	3.1, 3.5
Proakis and Salehi, *Communication Systems Engineering*	3.2
Rappaport, *Wireless Communications: Principles and Practice*	3.2
Richards, *Fundamentals of Radar Signal Processing*, 2nd ed.	3.3
Johnson and Graham, *High-Speed Digital Design*	3.4
Ott, *Electromagnetic Compatibility Engineering*	3.4, 4.6
Simon, *Optimal State Estimation*	3.5
Love, *Linux Kernel Development*, 3rd ed.	4.2
Gregg, *BPF Performance Tools*	4.2, 4.5
Madieu, *Linux Device Driver Development*, 2nd ed.	4.3
Bugnion, Nieh and Tsafrir, *Hardware and Software Support for Virtualization*	4.4

Free documents

Document	Chapters
TI, *Op Amps for Everyone* (SLOD006)	1.2
NXP UM10204, *I^2C-bus specification and user manual*	1.3, 2.6
TI SLAU723A, *MSP432E4 Technical Reference Manual*; SLAU748B, *MSP-EXP432E401Y LaunchPad user's guide*	2.6, 3.4, 3.5

Document	Chapters
Arm, *Cortex-M4 Devices Generic User Guide*	2.6
RISC-V Unprivileged and Privileged ISA specifications	2.3, 2.5
Intel 64 and IA-32 Software Developer's Manual, volume 3	2.5, 2.7, 4.1, 4.4
Nagarajan, Sorin, Hill and Wood, *A Primer on Memory Consistency and Cache Coherence*, 2nd ed.	2.4
Drepper, *What Every Programmer Should Know About Memory*	2.4
UEFI, ACPI, TCG PC Client Platform Firmware Profile and DMTF Redfish specifications	2.5, 2.7, 4.7
Arthur and Challener, *A Practical Guide to TPM 2.0* (Apress Open)	2.7, 4.6
IETF RFC 9334, *Remote Attestation Procedures Architecture*	2.7, 4.6, 4.7
Åström and Murray, *Feedback Systems*, 2nd ed.	3.1
TI TIDUCF1 (drone ESC reference design) and SPRUJ26 (Motor Control SDK lab guide)	3.1
TI SWRA588A (CC1312R LaunchPad) and SWCU185 (CC13x2 Technical Reference Manual)	3.2
Banerjee, *PLL Performance, Simulation, and Design* (TI)	3.3
TI SNAU217B (LMX2572EVM), SWRU596 (IWRL6432BOOST), SPYY005 and SWRA553 (mmWave fundamentals and chirp design)	3.3
OCP DC-SCM 2.0, Open Rack v3 and Yosemite v3 specifications	3.4, 4.6, 4.7
Raspberry Pi *Compute Module 5 Datasheet* and the CM5 IO Board design files	3.4
Arpaci-Dusseau, *Operating Systems: Three Easy Pieces*	4.1
Cox, Kaashoek and Morris, *xv6: a simple, Unix-like teaching operating system*	4.1
NVM Express Base, PCIe Transport, NVM Command Set and NVMe over Fabrics specifications	4.1, 4.3, 4.5
McKenney, *Is Parallel Programming Hard, And, If So, What Can You Do About It?*	4.2
Corbet, Rubini and Kroah-Hartman, *Linux Device Drivers*, 3rd ed.; Bootlin's kernel and driver training materials	4.2, 4.3
OASIS virtio 1.2 specification; Linux `Documentation/virt/kvm/api.rst`	4.4

Datasheets, application notes and user's guides for each part are linked from TI's product pages; download the ones a chapter names into a `datasheets/` directory in your labs repository, so your notes can cite page numbers that will not move.

PART 7

Progress tracker

The tables below list every lab in the course, in order. Date each line when the lab's done-when criteria all hold and its evidence is filed, and the chapter line when its completion checklist is ticked. The sign-off page at the end of each unit is the formal record, and this tracker shows where you are.

Unit 1: Circuits, Sensors and Signals

Lab	Title	Date done
Chapter 1.1 Electricity, components and measurement		
Lab 1.1.1	Divider, LED and source resistance	
Lab 1.1.2	RC charging and a threshold crossing	
Lab 1.1.3	The MOSFET as a switch	
Lab 1.1.4	Decoupling failure on the oscilloscope	
Checklist	*Every line of Chapter 1.1's completion checklist ticked*	
Chapter 1.2 The analogue signal chain		
Lab 1.2.1	Op-amp stages and their limits	
Lab 1.2.2	A second-order filter and its Bode plot	
Lab 1.2.3	Sampling, aliasing and quantisation with the ADS1115	
Lab 1.2.4	A load cell on a 24-bit converter	
Checklist	*Every line of Chapter 1.2's completion checklist ticked*	
Chapter 1.3 Sensors with TI silicon		
Lab 1.3.1	Bus bring-up and a driver from the datasheet	
Lab 1.3.2	Temperature and humidity metrology	
Lab 1.3.3	Power metrology with the INA228	
Lab 1.3.4	Magnetic angle with the TMAG5273	
Lab 1.3.5	Optical time of flight with the OPT3101	
Lab 1.3.6	An error budget for the drone's battery current	
Checklist	*Every line of Chapter 1.3's completion checklist ticked*	
Chapter 1.4 Power conversion and batteries		
Lab 1.4.1	Linear versus switching	
Lab 1.4.2	Inside a buck converter	
Lab 1.4.3	Protection: inrush, short circuit and reverse polarity	
Lab 1.4.4	Characterise the drone battery	
Lab 1.4.5	Two power trees	
Checklist	*Every line of Chapter 1.4's completion checklist ticked*	
Chapter 1.5 Signals on wires and RF fundamentals		
Lab 1.5.1	Reflections on a cable	
Lab 1.5.2	The network analyser and S-parameters	
Lab 1.5.3	Build and tune an antenna	
Lab 1.5.4	Using the spectrum analyser safely	
Lab 1.5.5	An FM receiver from scratch	
Lab 1.5.6	Link budgets for the drone	
Checklist	*Every line of Chapter 1.5's completion checklist ticked*	

Unit 2: Logic, Processors and Firmware

Lab	Title	Date done
Chapter 2.1 Digital logic		
Lab 2.1.1	Gates from transistors	
Lab 2.1.2	A 4-bit adder in SN74HC logic	
Lab 2.1.3	Flip-flops and a setup-time violation	
Lab 2.1.4	A finite state machine in gates	
Lab 2.1.5	A datapath in Digital	
Checklist	*Every line of Chapter 2.1's completion checklist ticked*	
Chapter 2.2 HDL and FPGA		
Lab 2.2.1	Toolchain, blink and a timing failure on purpose	
Lab 2.2.2	UART transmitter and receiver	
Lab 2.2.3	An asynchronous FIFO with a formal proof	
Lab 2.2.4	A memory-mapped peripheral	
Lab 2.2.5	The register map over SPI	
Checklist	*Every line of Chapter 2.2's completion checklist ticked*	
Chapter 2.3 Build a CPU		
Lab 2.3.1	Single-cycle RV32I	
Lab 2.3.2	A five-stage pipeline	
Lab 2.3.3	On the FPGA at speed	
Lab 2.3.4	Extensions	
Lab 2.3.5	Read three other cores	
Checklist	*Every line of Chapter 2.3's completion checklist ticked*	
Chapter 2.4 Memory and interconnect		
Lab 2.4.1	Caches on your CPU	
Lab 2.4.2	External SDRAM	
Lab 2.4.3	Memory experiments on a real machine	
Lab 2.4.4	PCIe by hand	
Lab 2.4.5	A page-table walker	
Checklist	*Every line of Chapter 2.4's completion checklist ticked*	
Chapter 2.5 Privilege and the platform		
Lab 2.5.1	M-mode on your CPU	
Lab 2.5.2	U-mode and protection on your CPU	
Lab 2.5.3	x86-64 bare metal in QEMU and on the server	
Lab 2.5.4	Read your server's ACPI and SMBIOS	
Checklist	*Every line of Chapter 2.5's completion checklist ticked*	
Chapter 2.6 Bare-metal firmware		
Lab 2.6.1	Blink from a blank directory	
Lab 2.6.2	Clocks, a proper UART and time	
Lab 2.6.3	I^2C and SPI masters: sensors, a fan and the FPGA	
Lab 2.6.4	DMA	
Lab 2.6.5	Faults and the watchdog	
Lab 2.6.6	A signed A/B bootloader	
Lab 2.6.7	A second core: the Cortex-M0+	
Checklist	*Every line of Chapter 2.6's completion checklist ticked*	
Chapter 2.7 Boot and platform firmware		
Lab 2.7.1	A boot sector, then a two-stage loader	
Lab 2.7.2	A UEFI loader	

Lab	Title	Date done
Lab 2.7.3	Secure Boot with your own keys	
Lab 2.7.4	Measured boot and remote attestation	
Lab 2.7.5	Arm boot chain and a device-tree overlay	
Lab 2.7.6	Zero-touch netboot	
Lab 2.7.7	BMC and Redfish, then OpenBMC	
Checklist	*Every line of Chapter 2.7's completion checklist ticked*	

Unit 3: Control, Radio and Boards

Lab	Title	Date done
Chapter 3.1 Motors, estimation and feedback control		
Lab 3.1.1	The thrust stand	
Lab 3.1.2	Field-oriented control from scratch	
Lab 3.1.3	Your FOC against a commercial ESC	
Lab 3.1.4	The IMU and its noise	
Lab 3.1.5	Attitude estimation	
Lab 3.1.6	Magnetometer calibration and heading	
Lab 3.1.7	PID on a one-axis rig	
Checklist	*Every line of Chapter 3.1's completion checklist ticked*	
Chapter 3.2 Radio links		
Lab 3.2.1	Bring-up and the SMA rework	
Lab 3.2.2	A packet radio below the stack	
Lab 3.2.3	Sensitivity and the packet-error waterfall	
Lab 3.2.4	The telemetry protocol	
Lab 3.2.5	Range in the field	
Lab 3.2.6	Demodulate your own packets	
Checklist	*Every line of Chapter 3.2's completion checklist ticked*	
Chapter 3.3 Microwave synthesis and FMCW radar		
Lab 3.3.1	A PLL you program	
Lab 3.3.2	Loop dynamics, spurs and lock time	
Lab 3.3.3	A chirp you can measure	
Lab 3.3.4	FMCW in simulation	
Lab 3.3.5	The IWRL6432 as an altimeter	
Lab 3.3.6	Raw ADC data	
Checklist	*Every line of Chapter 3.3's completion checklist ticked*	
Chapter 3.4 Board design and bring-up		
Lab 3.4.1	Specify four boards	
Lab 3.4.2	Board P: the drone's power module	
Lab 3.4.3	Board F: the flight-controller BoosterPack	
Lab 3.4.4	Board A: a BMC-lite on the MSP432E401Y	
Lab 3.4.5	Board B: a Compute Module 5 carrier	
Lab 3.4.6	Bring-up of Boards P and F	
Lab 3.4.7	Bring-up of Boards A and B	
Lab 3.4.8	The respin decision	
Checklist	*Every line of Chapter 3.4's completion checklist ticked*	
Chapter 3.5 The drone: integration, test and flight		
Lab 3.5.1	DShot over µDMA	
Lab 3.5.2	The receiver link	

Lab	Title	Date done
Lab 3.5.3	Flight software and simulation	
Lab 3.5.4	Assembly	
Lab 3.5.5	Rig tuning	
Lab 3.5.6	First hover	
Lab 3.5.7	Altitude hold with radar and barometer	
Lab 3.5.8	Telemetry and the flight report	
Checklist	*Every line of Chapter 3.5's completion checklist ticked*	

Unit 4: Systems Software and the Product

Lab	Title	Date done
Chapter 4.1 Write a kernel		
Lab 4.1.1	Boot, console and exceptions	
Lab 4.1.2	Memory	
Lab 4.1.3	Time, threads, scheduling and SMP	
Lab 4.1.4	User mode	
Lab 4.1.5	PCI, NVMe and a filesystem	
Lab 4.1.6	Choose one	
Checklist	*Every line of Chapter 4.1's completion checklist ticked*	
Chapter 4.2 Linux kernel internals		
Lab 4.2.1	Build and boot a kernel, then analyse a crash dump	
Lab 4.2.2	Tracing the kernel	
Lab 4.2.3	Modules, from hello to a character device under the sanitisers	
Lab 4.2.4	RCU in your module	
Lab 4.2.5	Scheduler and memory experiments	
Lab 4.2.6	Bisect a change, then a regression on hardware	
Lab 4.2.7	The node kernel and the patch process	
Checklist	*Every line of Chapter 4.2's completion checklist ticked*	
Chapter 4.3 Linux device drivers		
Lab 4.3.1	Mainline TI drivers by overlay	
Lab 4.3.2	An upstream-ready IIO driver for the OPT4048	
Lab 4.3.3	The ADS1220 over SPI: a data-ready trigger and a buffer	
Lab 4.3.4	Fan control: a hwmon driver, a thermal zone and PMBus	
Lab 4.3.5	Your FPGA over SPI, through regmap	
Lab 4.3.6	PCIe with MSI and DMA in QEMU	
Lab 4.3.7	NVMe, in the tree	
Lab 4.3.8	The 82599 in the kernel, then in userspace	
Checklist	*Every line of Chapter 4.3's completion checklist ticked*	
Chapter 4.4 Virtualisation and hypervisors		
Lab 4.4.1	A first guest through /dev/kvm	
Lab 4.4.2	Boot Linux with your own virtio-blk and virtio-net	
Lab 4.4.3	Your kernel as a guest, then a passed-through virtual function	
Lab 4.4.4	A bare-metal VT-x hypervisor	
Lab 4.4.5	Firecracker, Cloud Hypervisor and a VMM API for Compute	
Lab 4.4.6	Live migration and the multi-tenant node	
Lab 4.4.7	Arm, for comparison	
Checklist	*Every line of Chapter 4.4's completion checklist ticked*	
Chapter 4.5 Storage and network data paths		

Lab	Title	Date done
Lab 4.5.1	Measure the local storage path	
Lab 4.5.2	SPDK, and a vhost-user disk for your VMM	
Lab 4.5.3	Reed-Solomon in C++ at memory bandwidth	
Lab 4.5.4	NVMe over Fabrics between two nodes	
Lab 4.5.5	The network path: kernel, XDP, AF_XDP and DPDK	
Lab 4.5.6	A Blocks prototype	
Checklist	*Every line of Chapter 4.5's completion checklist ticked*	
Chapter 4.6 Product engineering		
Lab 4.6.1	The Module Zero specification	
Lab 4.6.2	The threat model	
Lab 4.6.3	The firmware update architecture	
Lab 4.6.4	The manufacturing and test plan	
Lab 4.6.5	The compliance plan	
Lab 4.6.6	The cost model	
Checklist	*Every line of Chapter 4.6's completion checklist ticked*	
Chapter 4.7 Capstone: Module Zero		
Lab 4.7.1	Week 1: node image and join	
Lab 4.7.2	Week 2: quorum and identity	
Lab 4.7.3	Week 3: compute	
Lab 4.7.4	Week 4: Blocks	
Lab 4.7.5	Week 5: updates and management	
Lab 4.7.6	Week 6: integration and recording	
Checklist	*Every line of Chapter 4.7's completion checklist ticked*	